Frameworx Standard

Information Framework (SID)

Customer Business Entities

Information Framework Suite
GB922 Customer
Release 16.5.1
February 2017

Latest Update: Frameworx Release 16.5	**TM Forum Approved** **Suitable for Conformance**
Version 16.5.2	**IPR Mode: RAND**

Notice

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1.1. Customer

1.1.1. Customer

Customers are at the center of any enterprise. Without customers at some point in its like, an enterprise cannot exist. Within the SID model, quite a bit of what is normally thought of as customer data resides within the Party business entity. This data includes attributes that describe individuals and organizations, such as name, address, phone, fax, email, and other contact information. More specifically, the role object model is used to abstract the notion of a "customer" into a specific subclass of PartyRole, which is documented in GB922, Addendum 1P. This is shown in Figure C.01 below.

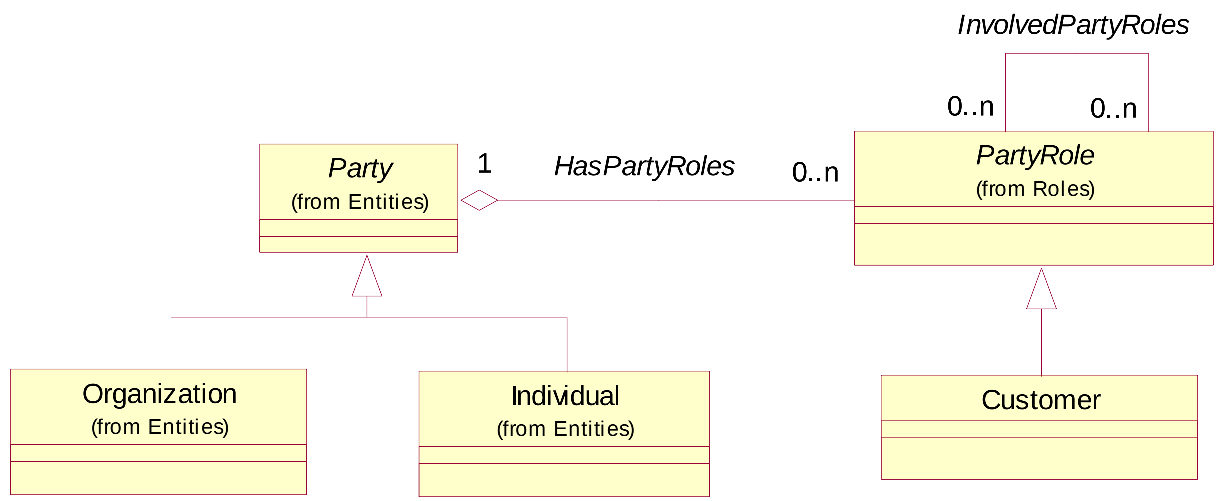

Figure C.01 – Customer as a Type of Party Role

This makes the Customer model inherently extensible. Customers can be conceptualized as an Individual, a group of people, or an organization. By making Customer a subclass of PartyRole, the SID avoids “hard-wiring” specific people or organizations as “customers”. Rather, the SID enables Customer to be one of possibly many roles played.

The primary business entities in the Customer Data model are Customer and Customer Account. The other business entities are shown for illustrative purposes and may be used as desired. Additional business entities, unique to a given enterprise, can be added to extend this model.

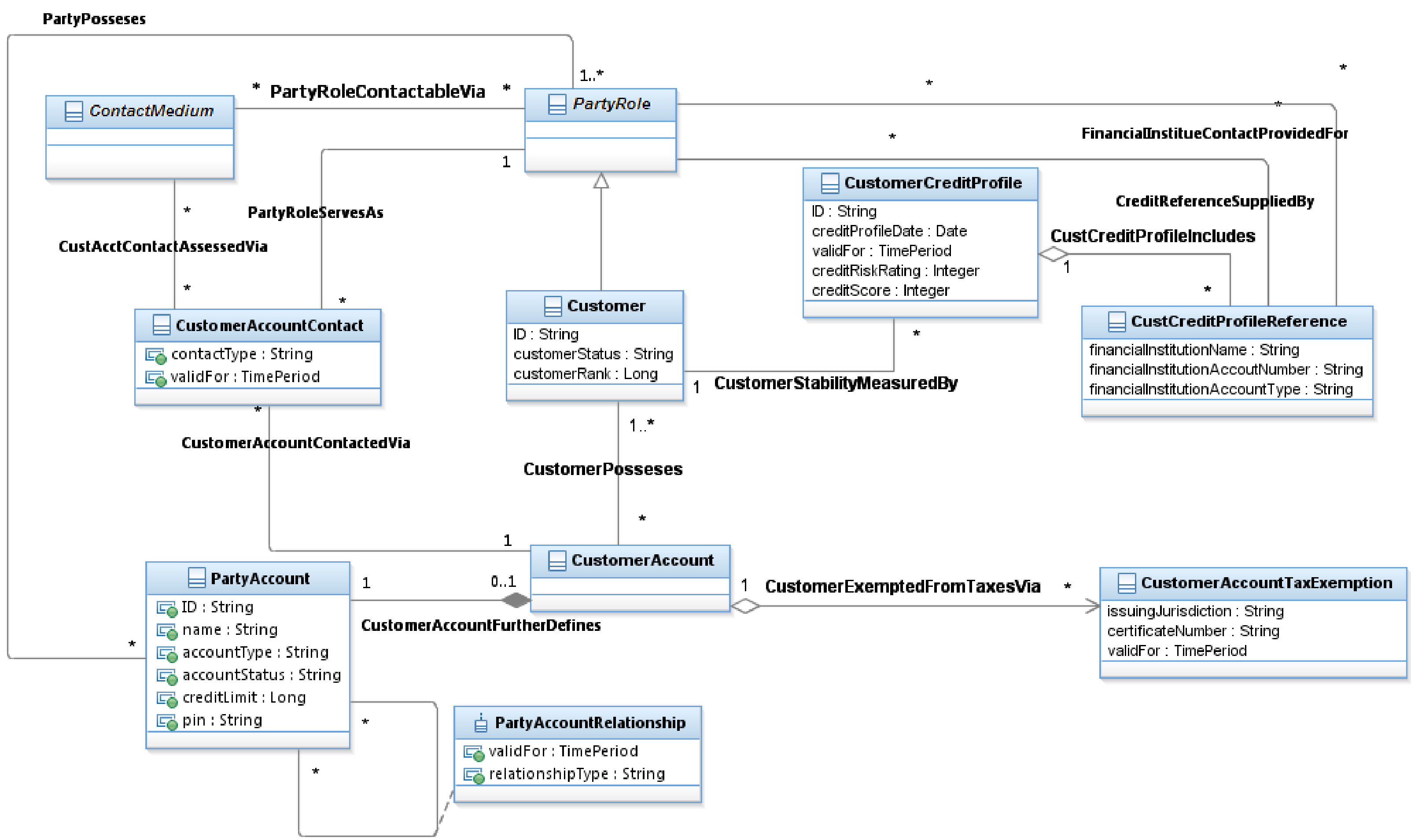

Figure C.02 – Customer Business Entities

1.2. Customer Order

1.2.1. Customer Order

Note: Some but not all enterprises consider an Order to be a type of an Agreement. In the SID model, a PartyOrder is considered to be a type of Request (which is a type of BusinessInteraction – see Addendum 1-BI Business Interaction) without the legal formalisms that characterize an Agreement. From a SID model perspective, a PartyOrder can be formalized by an Agreement and the Agreements relationship between the Agreement and the PartyOrder. This philosophy has the advantage of clearly separating two different concepts: a request (that is, the actual Party Order) from the legal formalisms associated with the request.

A Customer might place orders with the Service Provider. This is represented by the CustomerOrder.

CustomerOrder / CustomerOrderItem further define PartyOrder / PartyOrderItem. For further details about PartyOrder / PartyOrderItem refer to the EngagedParty guide book.

A PartyOrder is a type of BusinessInteration that represents a communication used to procure or update one or many Products in the context of a ProductOffering through all its PartyOrderItems.

The particularity of the CustomerOrder is to procure or update Products for Customer even if the CustomerOrder might be placed by the Service Provider when applying precautionary measures in case of bad debt.

One or many PartyRoles might be involved in a CustomerOrder / CustomerOrderItem such as Distributor, Holder, Buyer, Product User or DeliveryDriver.

The CustomerOrder and CustomerOrderItems specify the CustomerAccount to which the Products' are charged.

A complete CustomerOrder or CustomerOrderItems might be paid through one or many CustomerPayments.

A CustomerOrderItem may need other CustomerOrderItems to be completed; this dependency is represented through the "/CustomerOrderItemRefrencesReferencedBy" relationship. For example, using this relationship you can specify that CustomerOrderItem "A" requires CustomerOrderItem "B".

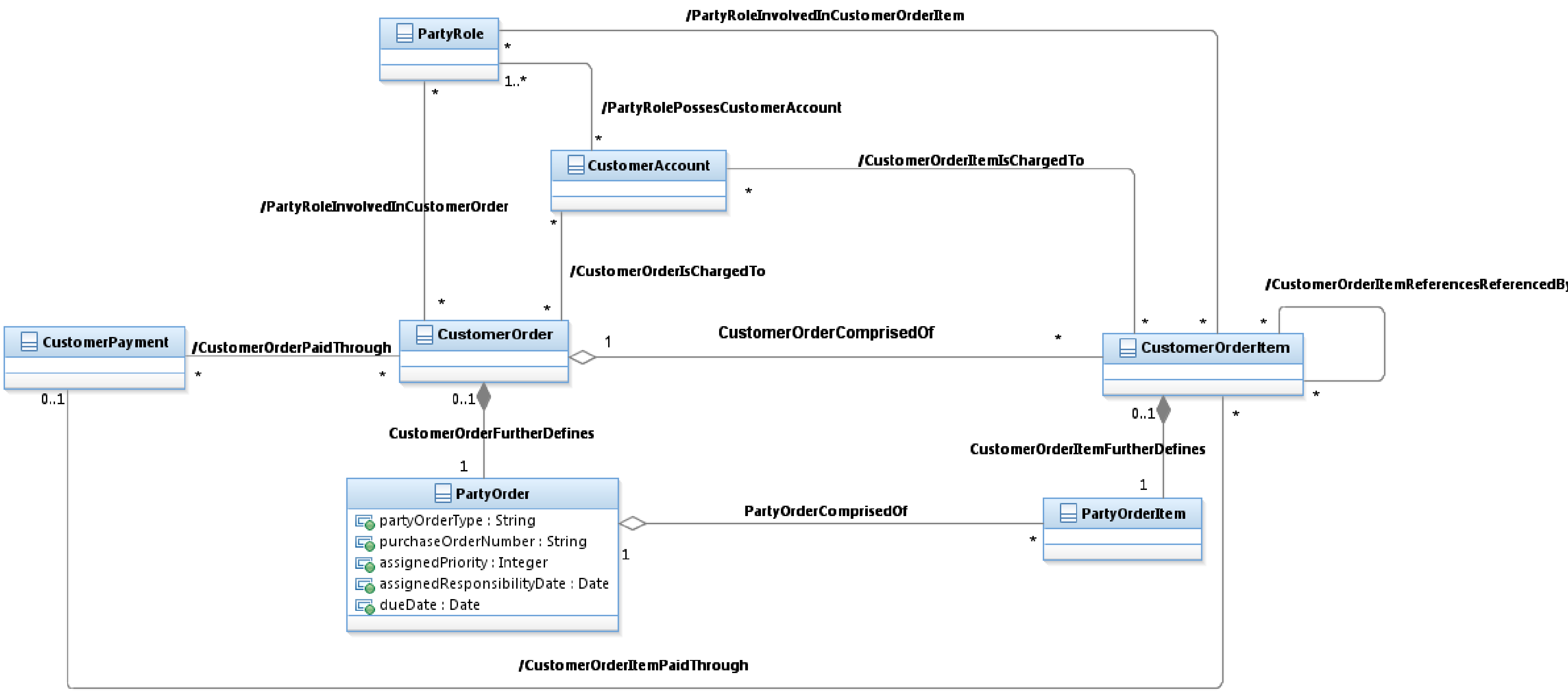

Figure C.03 – Customer Order Business Entities

1.2.2. CustomerOrderItem

Each CustomerOrderItem requires an action (AllowedProductAction) on a ProductSpecification or a ProductOffering. The configuration required is described by the related Product even if it isn't delivered yet.
To deliver the corresponding Product, the Place to deliver and Appointment(s) might be needed. An Appointment is an arrangement to do something or meet someone at a particular time and location.
A CustomerOrderItem may also specify the commitment chosen by the Customer (i.e. CommitmentTermOrCondition).

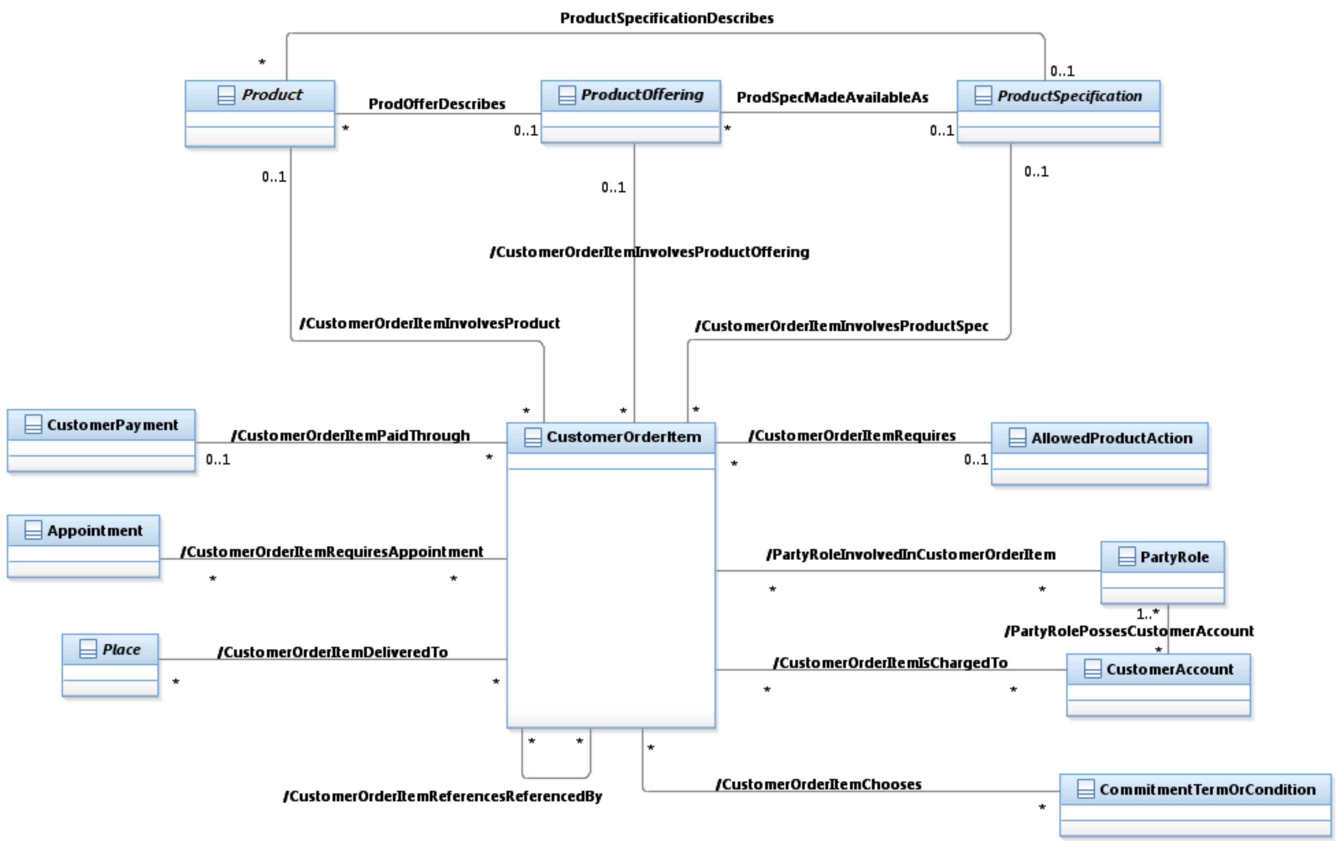

Figure C.03a – Customer Order Item Business Entities

1.3. Customer Service Level Agreement

1.3.1. Customer Service Level Agreement

A Customer Service Level Agreement differs from other types of Service Level Agreements due to the fact that a Customer is one of the PartyRoles involved in the Service Level Agreement, while other roles are involved in other types of agreements. For example, a Supplier is involved in a Supplier Service Level Agreement. See GB922 Addendum 1A for details of Agreements and Service Level Agreements. See GB922 Addendum 4SO for details of Service Level Specifications and Service Level Objectives.

The figure below provides an overview of a Service Level Agreement. It does not show all of the relevant associations to reduce the complexity of the figure. In particular, associations that BusinessInteraction or BusinessInteractionItems have with other entities are suppressed. Two examples of this are the suppression of the association between a BusinessInteractionItem and a ProductOffering as well as the association between a BusinessInteraction and a Customer playing a PartyRole.

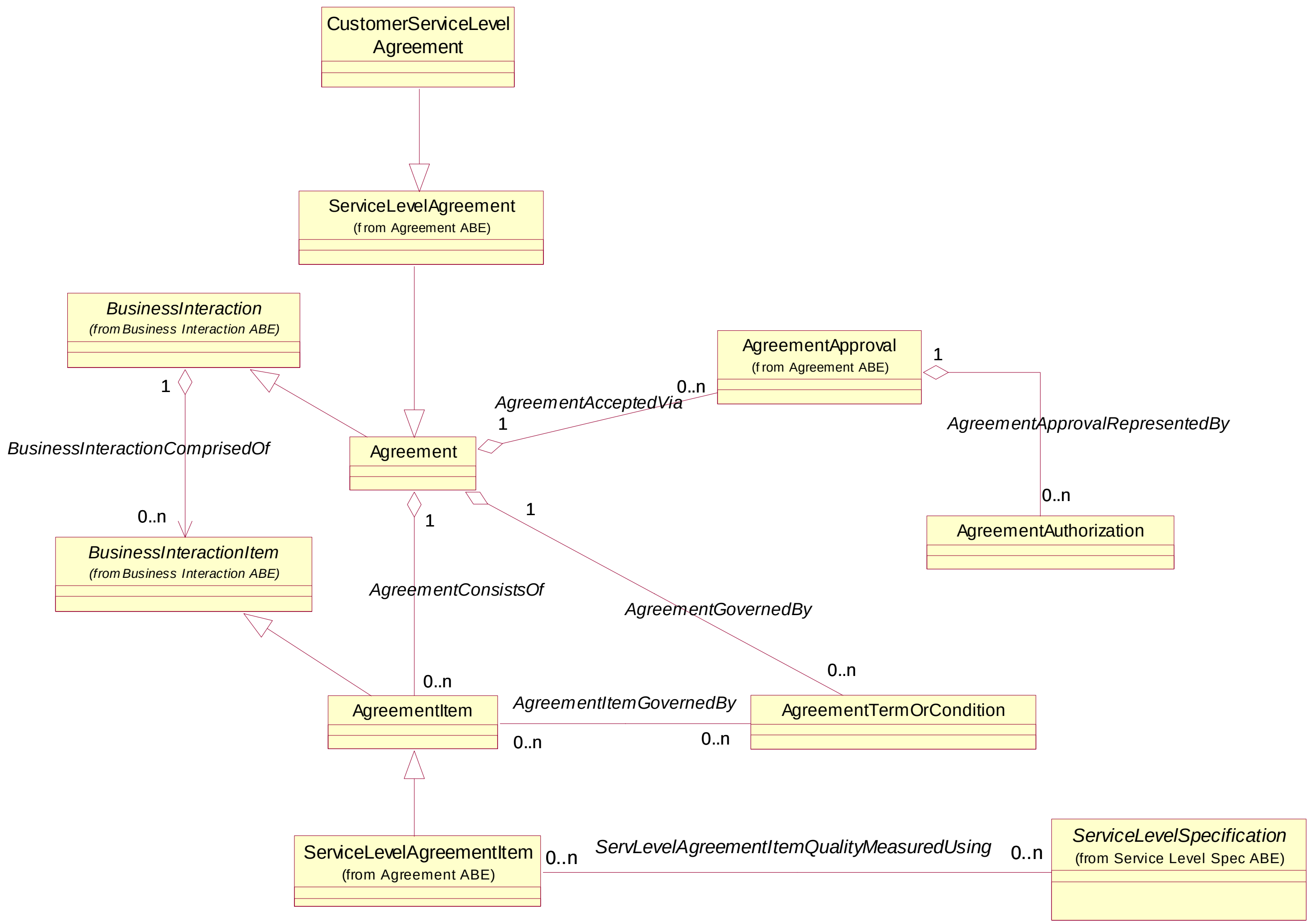

Figure C.04 – Customer Service Level Agreement Business Entities

1.4. Customer Interaction

The Customer Interaction Aggregate Business Entity has not been fully detailed at the time of this writing. Currently, there are six business entities in the model as shown in the figure below. Customer interactions take the form of requests (CustomerInquiry, CustomerBillInquiry, and CustomerInvoiceInquiry) and responses (CustomerQuoteOrOffer).

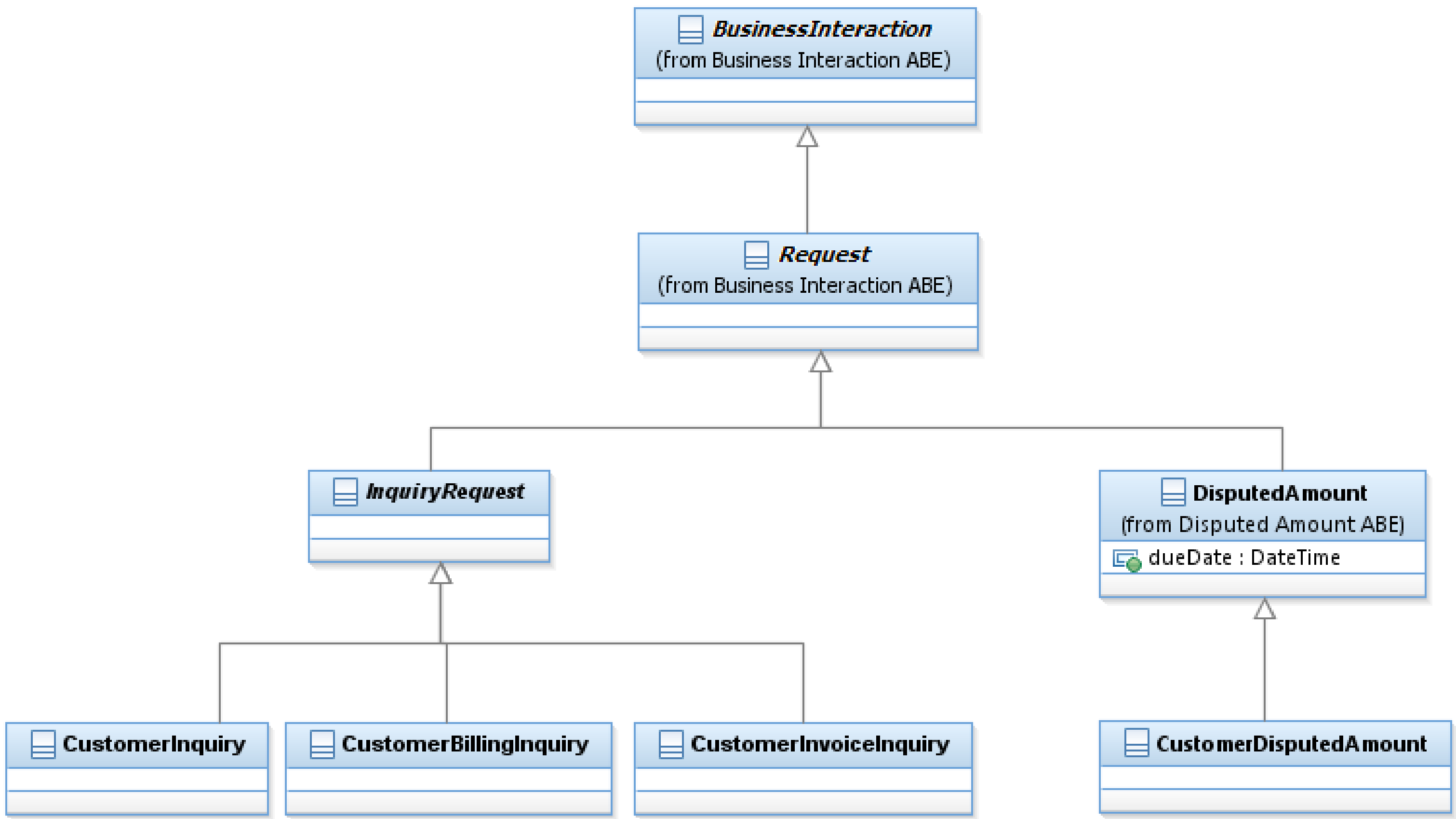

Figure C.05 – Customer Interaction: Types of Requests

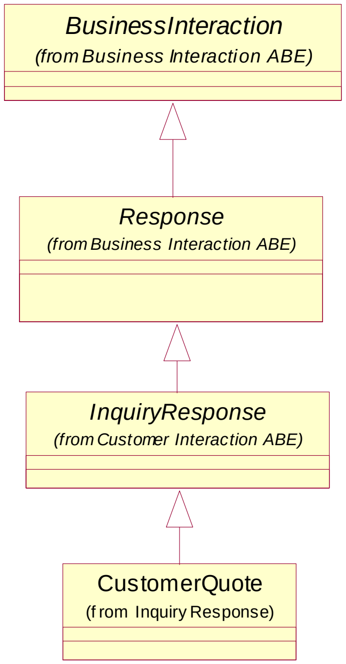

Figure C.06 – Customer Interaction: Types of Responses

1.4.1. References

IPDR	Service Specification Design Guide, Version 3.5.1
Jane M. Hunter	Telecommunications Billing Systems, ISBN 0-07-140857-6

1.4.2. CustomerDisputedAmount

CustomerDisputedAmount is a subclass of DisputedAmount which models amount in dispute between the enterprise and a customer (amount can be in dispute between the enterprise and other entities such as supplier or partner). The Customer can request a refund for a specific Charge (for example, because the service was not available in a specific time), for a bill (for example when a bill is too high because of unexpected charges such as data roaming) or by generic complaint (for example when the customer waited at home for a technician which did not arrive on time). Each such a request generates a CustomerDisputedAmount request which is associates optionally to the entity which is in dispute. The generic dispute is associated with the CustomerAccount.

In addition justified dispute generates AppliedCustomerBillingCredit for the justified amount. The association DisputeAppliesTo is required for accessing all CustomerDisputedAmounts which relate for specific CustomerAccount (for example when evaluating if the CustomerAccount should be moved to Collection handling)

All this is modeled in the following figure:

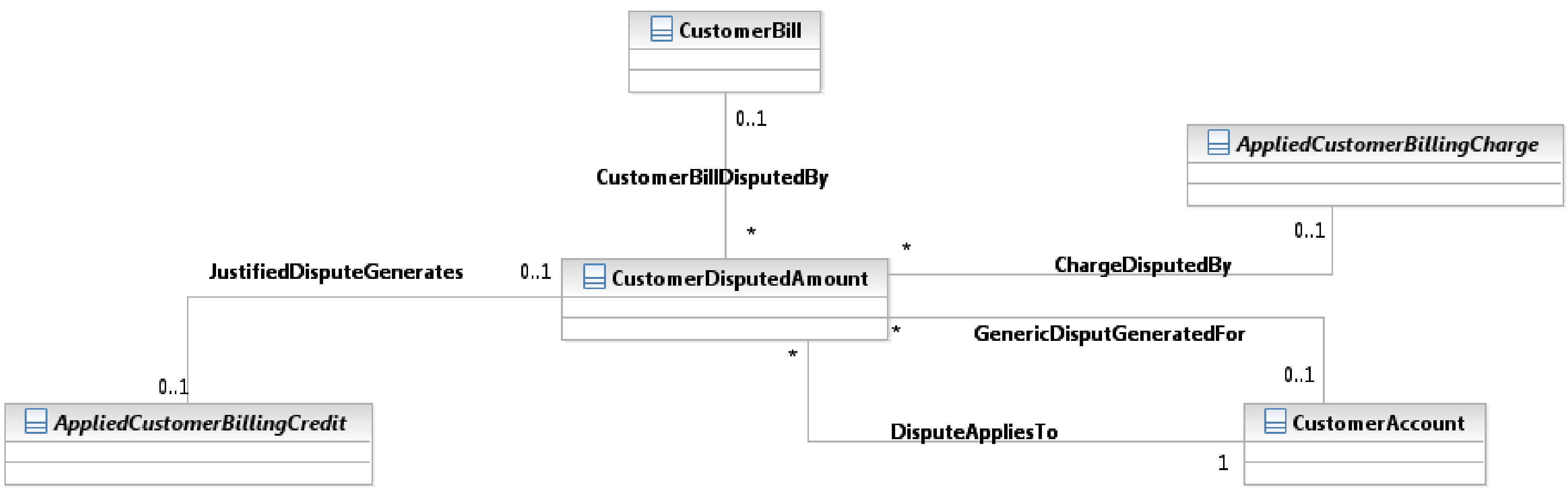

Figure C.07 – CustomerDisputedAmount Associations to Other Business Entities

1.5. Customer Problem

CustomerProblem is a type of business interaction which represents a problem which affects the customer. Such a problem can be a problem with the customer products (such as low quality of service), customer bills (such as unexpected high bill) or any other problem which influences the customer experience. CustomerProblem can be raised by the customer who noticed a problem, or by the CSP (using some analytics tools) which detects a problem and proactively work on solving it before the customer complains.

CustomerProblems may be nested for 2 business reasons:

- For attaching root cause problem to multiple reported *CustomerProblems* (typically related to multiple customers)
- For breaking a *CustomerProblem* to sub problems (typically when customer reports many problems during one call).

CustomerProblem may involve multiple parties, some of them are known PartyRoles (such as Customers) and some play a temporary role just for this problem (such as the reporter of the problem). For the latter cases the CSP can choose whether to create a PartyRole for a problem reporter or just use PartyInvolvementRole as association to Party rather than to PartyRole.

This part of the model is described in the following diagram:

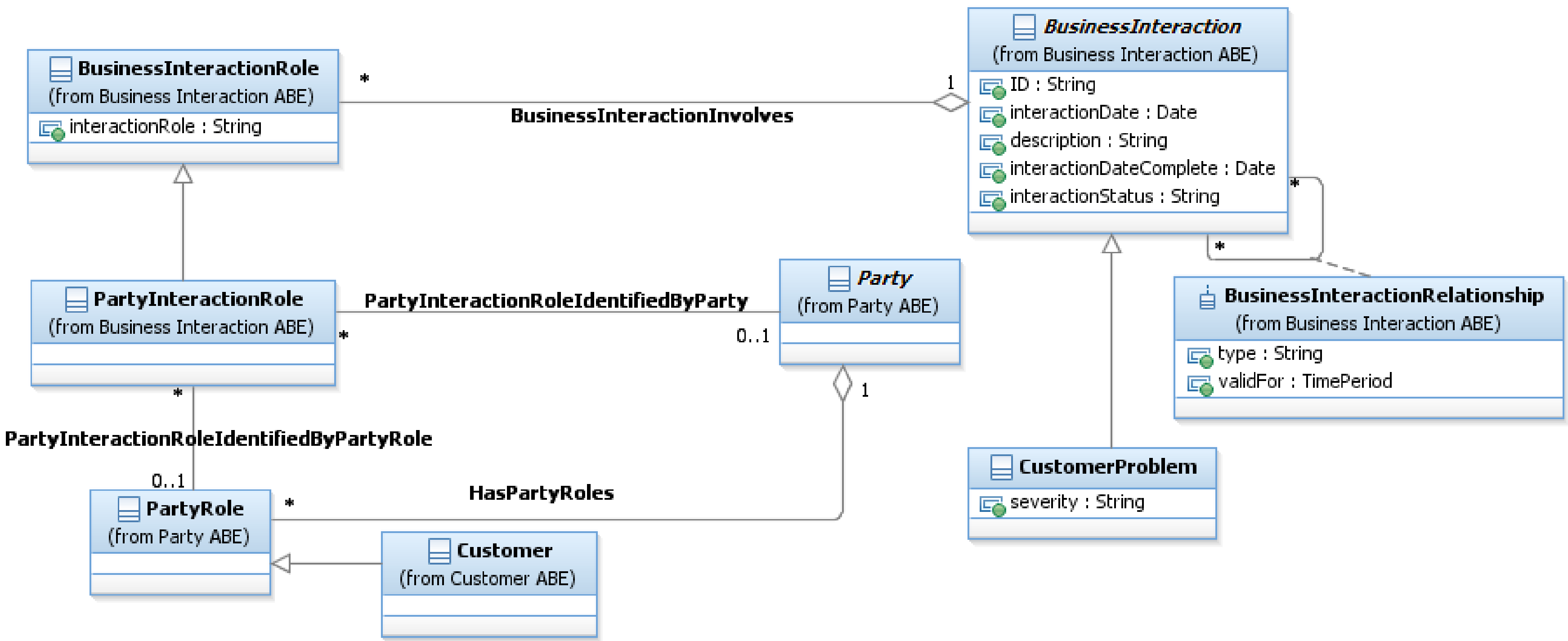

Figure C.08 – Customer Problem Context

A CustomerProblem may be associated with CustomerProblemTask(s) which represent a trackable task delegated from the CustomerProblem with a specified due date. CustomerProblem may also be an instance of a problem known to the service provider, optionally with a workaround to bypass this problem, in such a case, a KnownProblemDescription may be associated with the CustomerProblem.

A CustomerProblem is also optionally associated with Attachment (through inheritance from BusinessInteraction), typically such as Attachment represents a document (such as screenshot or diagnostics report) that the Customer provides that depicts the problem he encounters.

When a CustomerProblem is closed a CloseCustomerProblemSummary entity which records the closure activity is created. A CustomerProblem may associate to more than one CloseCustomerProblemSummary since a problem can be closed and then reopened (typically by the customer) if the problem is not solved.

The following diagram shows the CustomerProblem model:

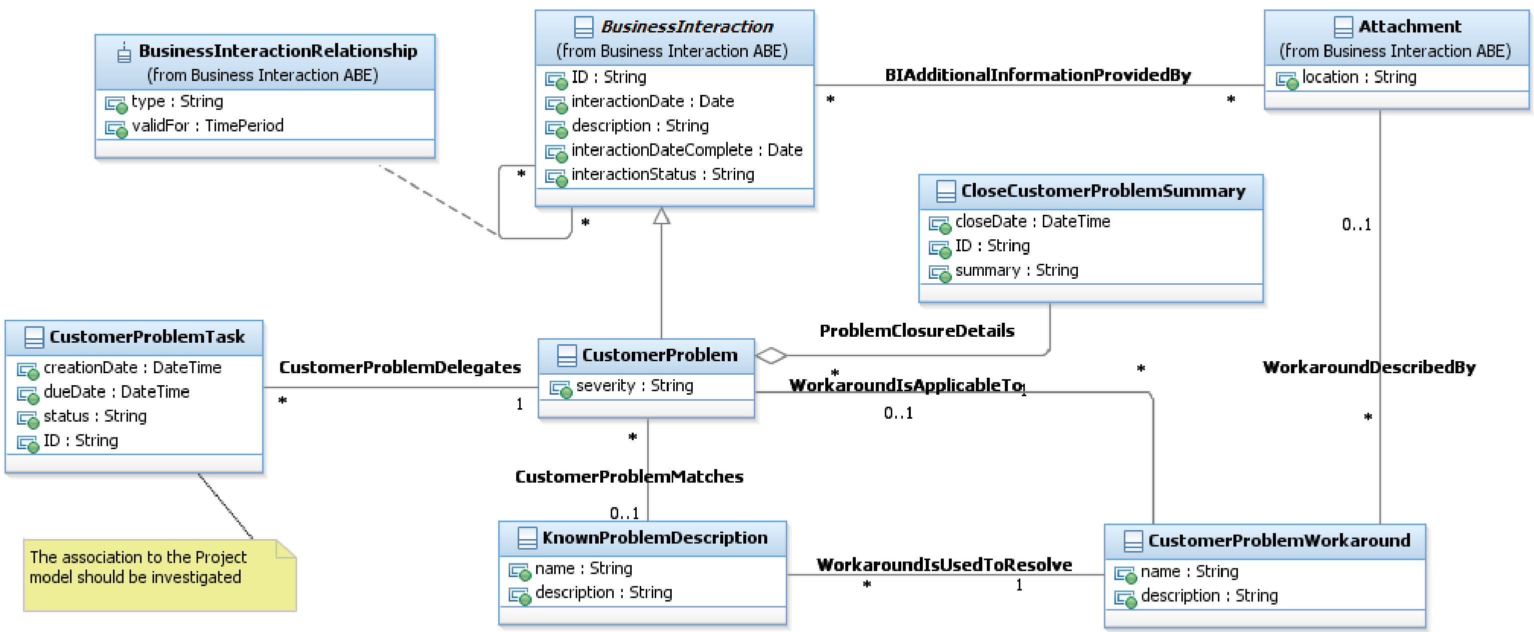

Figure C.09 – Customer Problem Model

The full CustomerModel (context and ABE model) is depicted in the diagram below:

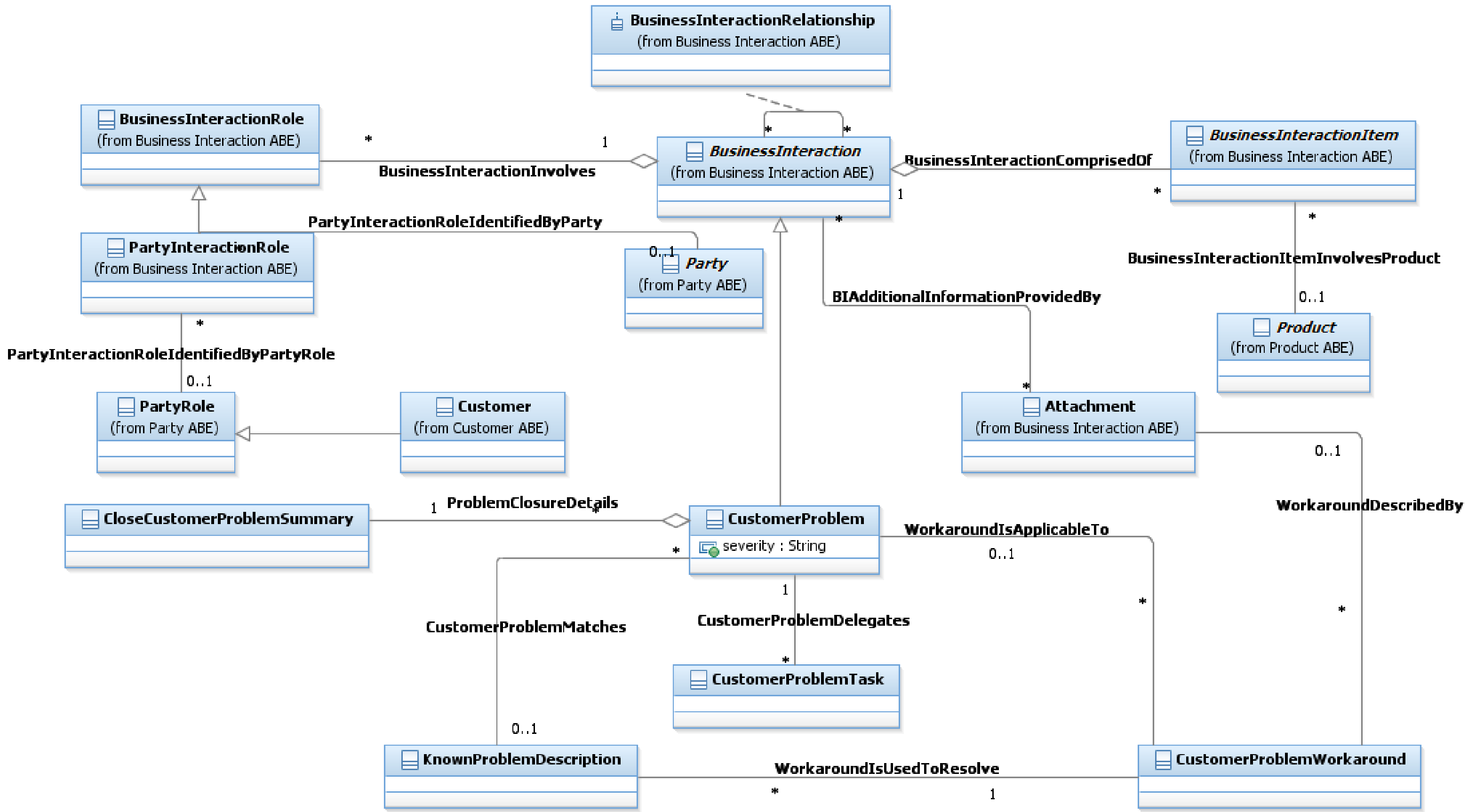

Figure C.10 – Customer Problem Full Model

1.6. Customer Billing

Customer Billing covers business entities utilized by rating and billing processes. These entities form the Applied Customer Billing Rate ABE, the Customer Bill ABE and the Customer Bill Collection ABE. Products are rated at different prices depending on ProductOfferingPrices, ProductOfferingPriceRules and additional terms & conditions that are determined by the CustomerAccount. Rating process takes ProductUsages on input and applies rates to them. The Billing process applies additional charges (for example, one time charges, recurring charges), allowances, discounts and taxes, to Products, then aggregates applied rates into bills and sends them to customers. With the emergence of real-time billing the boundary between rating and billing has become blurry.

1.6.1. Introduction

This document is intended to specify an entity model used in mediation, rating and billing processes.

Scope and Limitations

Hence the model:

- Has no distinction between prepaid and postpaid
- Has no distinction between an Invoice and a Bill
- Some financial entities that represent the integration with other ABEs such as Accounts Receivable and General Ledger are missing.

These issues will be addressed in following versions of the SID.

A typical billing process involves collecting business transaction details from network equipment, application servers and various other sources, correlating and formatting collected transaction details into billable events that a billing system can understand, guiding (associating usage to customer products), assigning charges to each billable event, applying other fees and discounts, applying taxes, and aggregating everything into customer bills (invoices) and receiving and recording payments from customers.

This model is not focused on any specific business domain but rather tries to build a model framework into which models for different billing domains would fit (for example, telecom billing, utilities billing). It also supports all three basic billing types: postpaid periodical billing, postpaid real-time billing and prepaid real-time billing. The model, by no means, represents the complete set of bill elements that can be incurred. The model provides a framework into which other types of bill elements can easily fit.

The overview figure below shows a simplified overview of the main billing domain business entities. The text following the figure provides an explanation of each of the figure's entities.

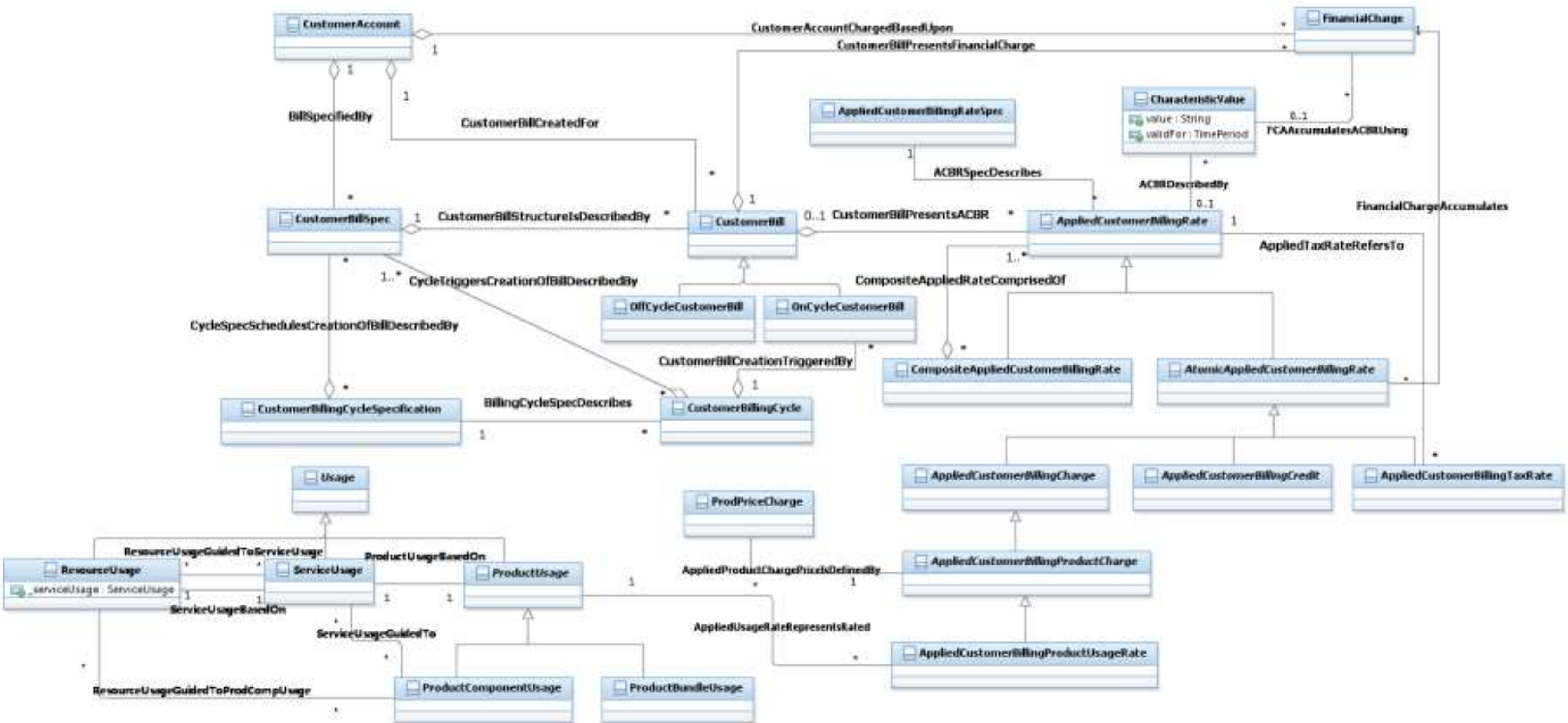

Figure C.11 – Billing Entities Overview

1.6.2. Applied Customer Billing Rate

The AppliedCustomerBillingRate business entity represents an applied billing rate assigned to the CustomerAccount. AppliedCustomerBillingRates are created before or during the billing cycle. The billing process aggregates applied rates into FinancialCharges which in turn are aggregated into one or more bills. Each applied billing rate is either atomic or composed of other applied rates. This is very useful for displaying AppliedCustomerBillingRates in the CustomerBill. For Financial purposes, however, the composite/atomic pattern is inappropriate since the AppliedCustomerBillingRate may be aggregated more than once (see example in Figure CE.01) and therefore FinancialCharge is presented which aggregates each AtomicAppliedCustomerBillingRate exactly once for reporting to the financial systems.

AppliedCustomerBillingRate is described by AppliedCustomerBillingRateSpec, and it is associated with dynamic Characteristics. This is due to the fact that each type of AppliedCustomerBillingRate may carry different Characteristics based on service type, line of business or other parameters.

The AtomicAppliedCustomerBillingRate class has four subclasses:

AppliedCustomerBillingCharge, that represents any kind of charge (except taxation charges);

AppliedCustomerBillingCredit, that represents any kind of credit;

AppliedCustomerBillingTaxRate, that represents a taxation charge;

AppliedCustomerPenaltyCharge that represents penalty charges such as late fees, payment rejection fees…

The AppliedCustomerBillingProductCharge is a subclass of the AppliedCustomerBillingCharge and represents a charge applied to a product, while its subclass AppliedCustomerBillingProductUsageRate represents the rate for a rated product usage event typically produced by a rating engine.

Examples of the AppliedCustomerBillingProductCharge are:

A recurring monthly fee charged for the voice service product.

A one-time installation fee charged when the customer subscribes for a voice service product.

Examples of the AppliedCustomerBillingProductUsageRate are:

When a customer makes a voice call a ServiceUsage instance representing call details is created. Afterwards the guiding process creates a ProductUsage instance that represents the ServiceUsage associated with the proper Product. Then the rating process applies a usage price to the ProductUsage instance and creates an AppliedCustomerBillingProductUsageRate instance representing the price of the rated Product Usage.

Similarly when a mobile service customer send an SMS, an AppliedCustomerBillingProductUsageRate representing charge for SMS transportation is created.

The AppliedCustomerBillingCredit business entity is an abstraction of an applied billing credit. The applied billing credit is usually created by the rating or billing process. Its creation can be governed by ProductPriceRules.

The AppliedCustomerBillingRebate is a subclass of AppliedCustomerBillingCredit and represents a rebate given to the customer account.

The AppliedCustomerBillingProductAlteration is a subclass of AppliedCustomerBillingCredit that represents a modification of the referred applied billing charge (either atomic or composite).

The AppliedCustomerBillingTaxRate represents taxes applied to billing charges. The entity may be linked to another applied rate (either atomic charge or composite rate) to indicate which applied rate it refers to. It is calculated during a billing process.

The CompositeAppliedCustomerBillingRate represents a sum of one or more other applied rates. It is formed by aggregating other AppliedCustomerBillingRates that may be either Composite or Atomic AppliedCustomerBillingRates. The main purpose of the CompositeAppliedCustomerBillingRate is to represent an aggregated view of comprising AppliedCustomerBillingRates that can be referenced by other AppliedCustomerBillingRates, for example, by AppliedCustomerBillingTaxRate.

The presented model by no means represents the complete set of applied billing rates that can be incurred. The model provides a framework into which other types of applied billing rates can easily fit.

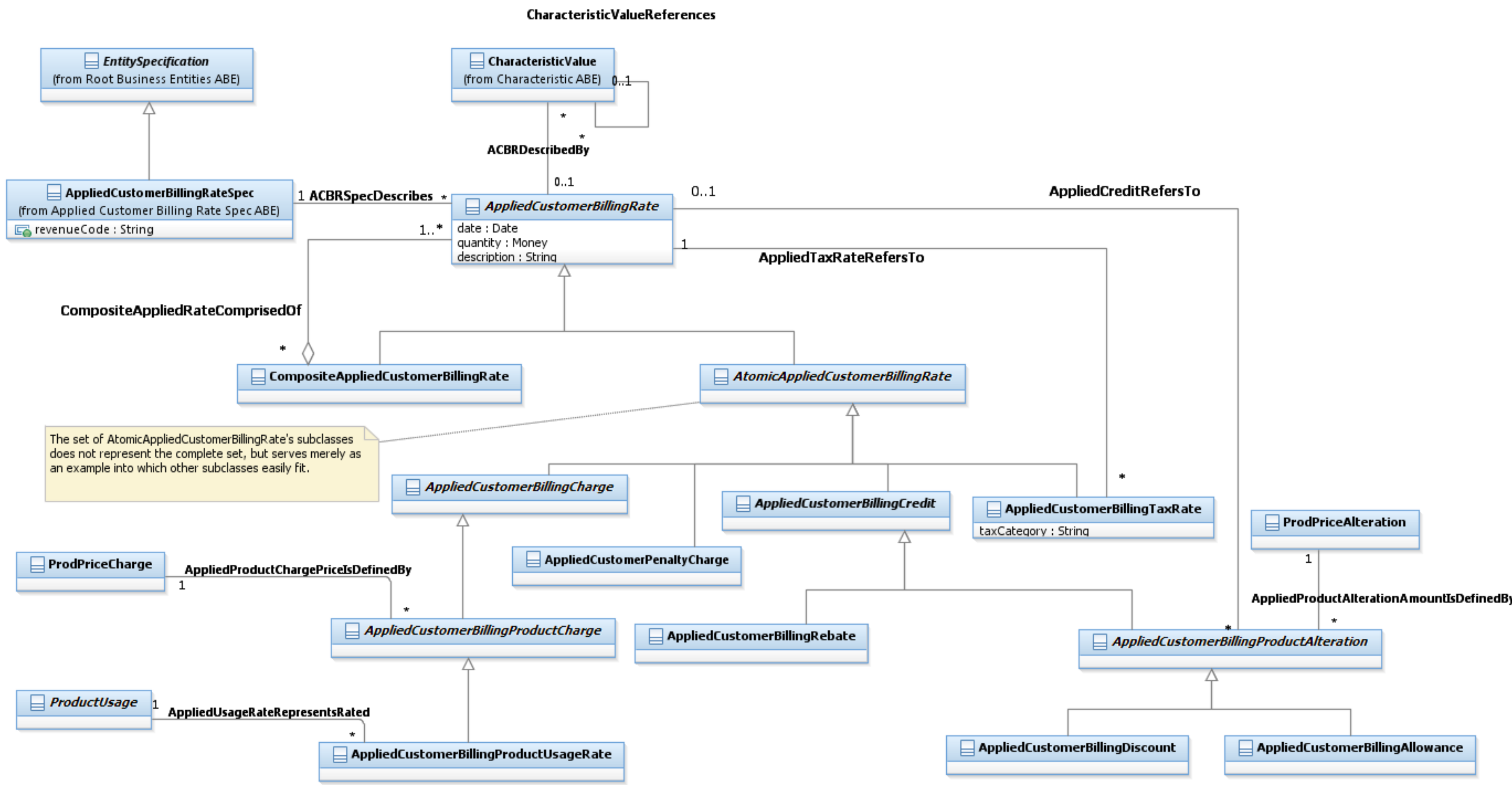

Figure C.12 – Applied Customer Billing Rate

1.6.3. Customer Bill

The ultimate goal of any billing process is creation of a bill (an invoice). The bill is created during a customer account billing cycle and associated with a customer account.

For financial reasons it is critical that each AtomicAppliedCustomerBillingRate amount appears exactly once on each CustomerBill. This is not well supported by the atomic/Composite pattern as explained in the Figure below:

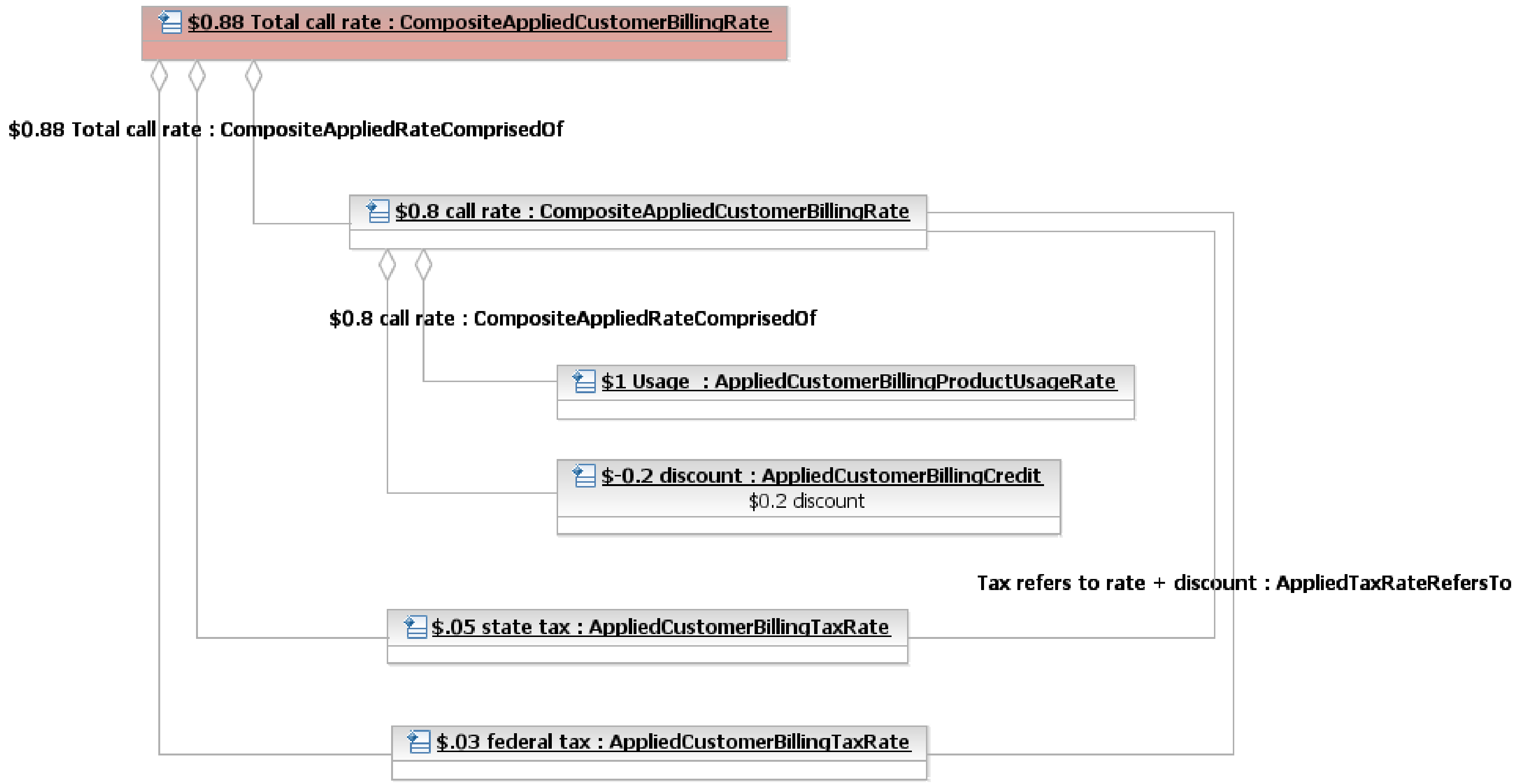

Figure CE.01 – Issues with Using Composite/Atomic Charges to Model Financial Reporting

For financial reasons we must report $0.88, however summing all these instances together gives us $2.56. If we sum, however, only the leaves of the tree (the AtomicAppliedCustomerBillingRates) we get exactly $0.88 ($1 - $0.2 + $.05 + $.03). Therefore the FinancialCharge business entity aggregates only AtomicAppliedCustomerBillingRate

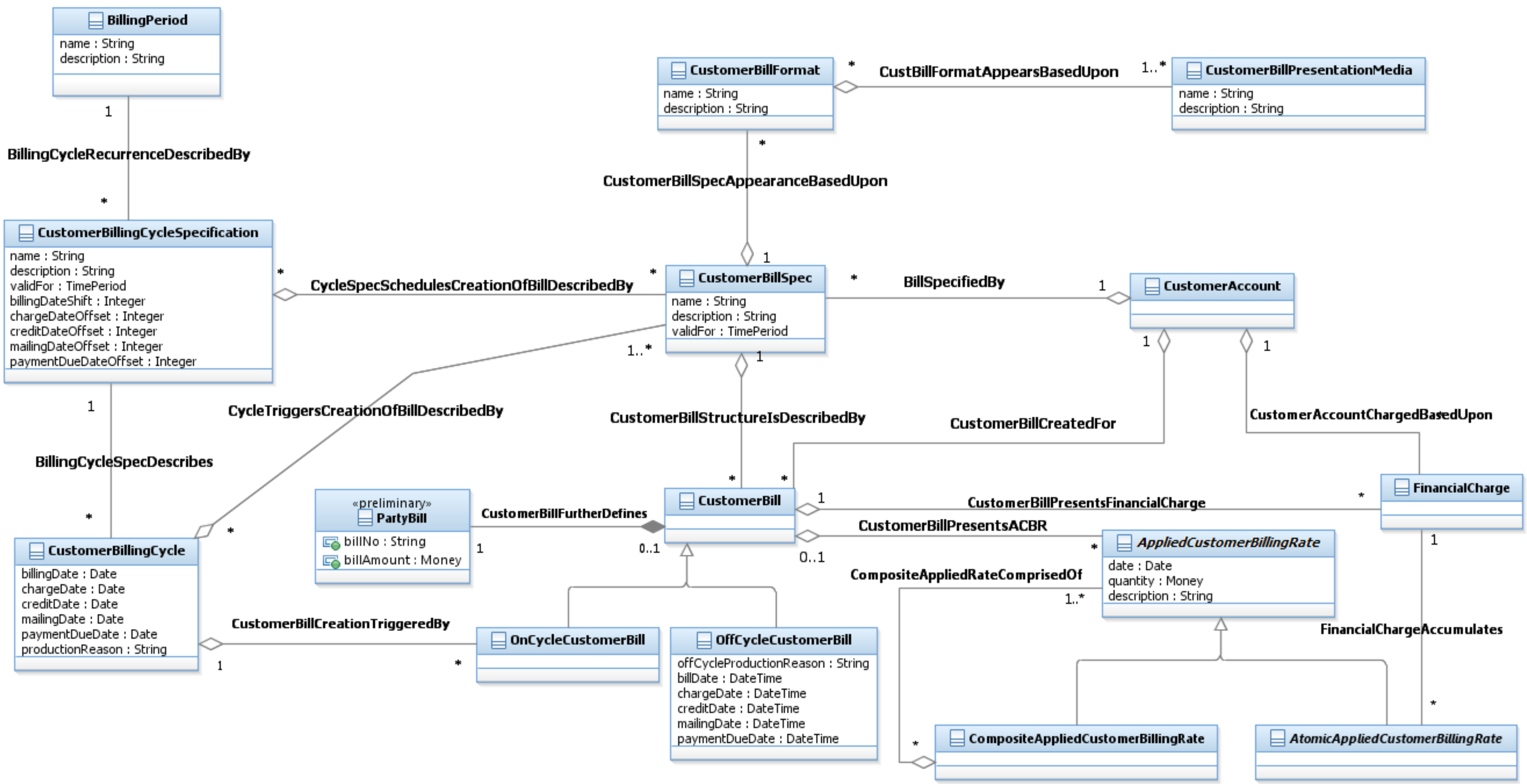

Figure C.13 – Customer Bill

The CustomerBillSpec describes the detailed structure of the customer's bill. It describes which AppliedCustomerBillingRates should be included into the bill, when to initiate the bill creation and bill presentation formats. The presentation format of the bill and its aggregated AppliedCustomerBillingRates are described by the CustomerBillFormat. The bill may be presented via various presentation media (for example, email, post mail, web page).

The CustomerBillingCycleSpecification, as shown in the figure above, identifies when to initiate a billing cycle and various sub steps of a billing cycle. It defines a date to be shown on the bill, the date through which charges and credits previously received by the billing system will appear on the bill, the mailing date and the payment due date.

The CustomerBillingCycle entity represents a particular occurrence of a billing cycle. Besides significant dates, it records who (or what) triggered the cycle. The CustomerBillingCycle aggregates all CustomerBillSpecs and in this way defines, which bills are to be created during the represented billing cycle.

CustomerBills can be created as a result of a cycle run or as a result of other events such as customer request or account close. The bills which are created as a result of a cycle run are described by the entity OnCycleCustomerBill (which is a subclass of CustomerBill) which includes a mandatory association to the CustomerBillingCycle that triggered the cycle run. Bills which are generated as a result of a different event are represented by the entity OffCycleCustomerBill ().

The CustomerBill entity is used to model a bill (an invoice). It represents a total amount due for all products during the billing period and all significant dates (i.e. billDate, chargeDate, creditDate, mailingDate and paymentDueDate). The structure of the bill and its presentation formats are described by the CustomerBillSpec. OnCycleCustomerBill is generated during the CustomerBillingCycle and is associated with a customer account. The CustomerBill aggregates FinancialCharges and displays AppliedCustomerBillingRates.

Let's imagine the AtomicAppliedCustomerBillingRate and CompositeAppliedCustomerBillingRate, the latter containing the former. While both applied rates may be included into the bill, only one of both may be summarized into the bill amount. The typical reason to include CompositeAppliedCustomerBillingRate into the bill is to present a partial sum, either for customer information or because some other applied rates refer to it. FinancialCharge, on the other hand, is the entity that accumulates AtomicAppliedCustomerBillingRate into the bill amount, and therefore the entity which influences the customer account's balance. The accumulation is based on characteristic values of AppliedCustomerBillingRate as described in the figure below:

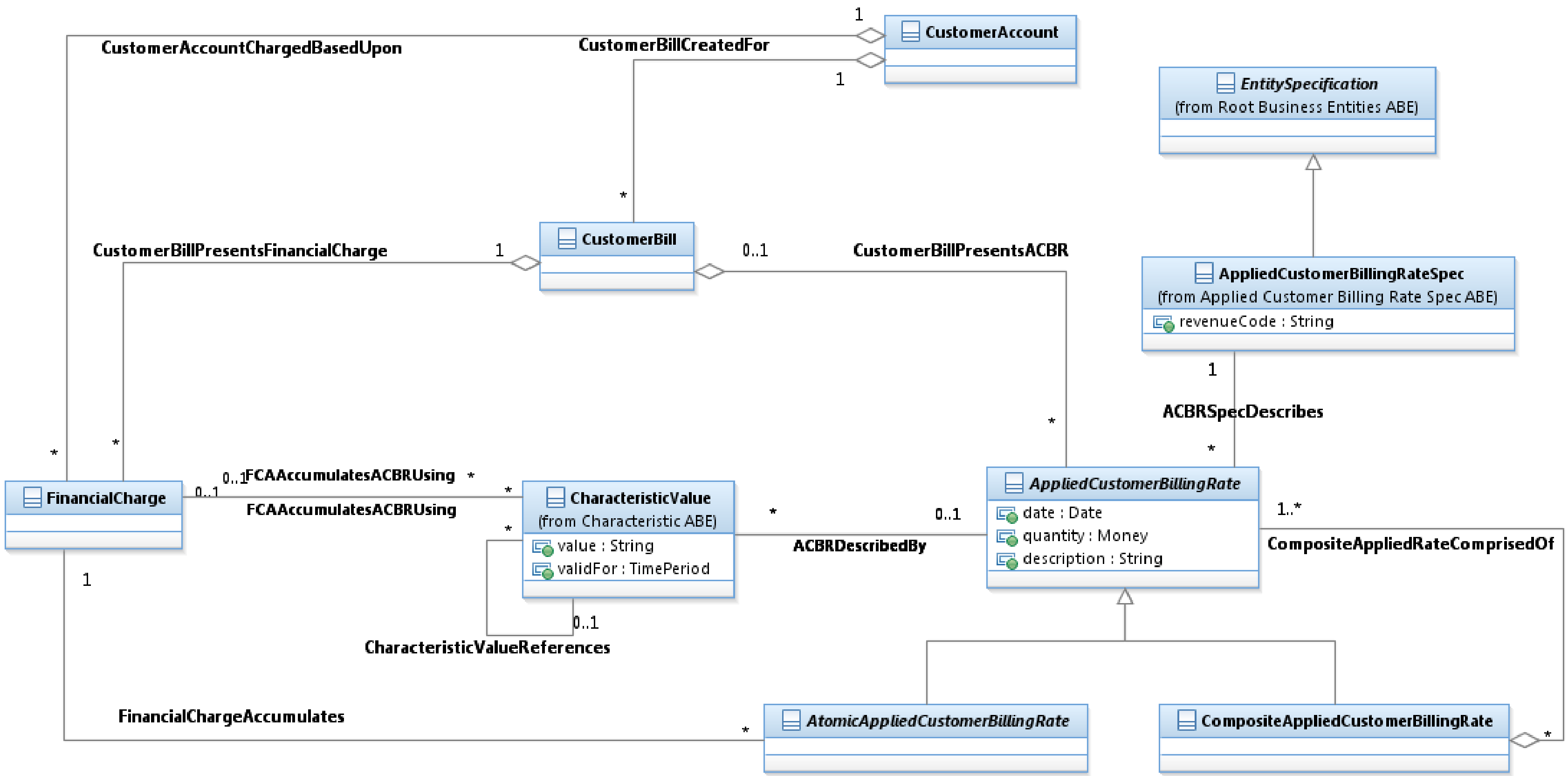

Figure C.14 – FinancialCharge Accumulates AtomicAppliedCustomerBillingRate

In this figure we see that the CustomerBill presents both AppliedCustomerBillingRates and FinancialCharges – this depends on the business requirements, configuration and regulations – one or both options are possible pending the CSP business.

The important thing (and change from Information Framework versions prior to version 12) is that the CustomerAccount now is based upon the FinancialCharges and not upon the AppliedCustomerBillingRate, for reasons which are explained above. FinancialCharge can accumulate AppliedCustomerBillingRates based on its characteristic values, so that it can accumulate based on ACBR type, line of business, service type or any other ACBR characteristic.

The presented model by no means represents the complete set of bill elements that can be incurred. The model provides a framework into which other types of bill elements can easily fit.

1.6.4. Customer Billing Statistic

The CustomerBillingStatistic and related business entities are used to model various totals and other statistical data associated with a customer account and its products that are of billing process interest over a certain period of time. For example, they can represent customer account total annual revenue or total volume of product usage per quarter. Application of various billing credits (e.g. discount, credit) can be triggered by billing statistic data meeting (for example, exceeding) a certain value.

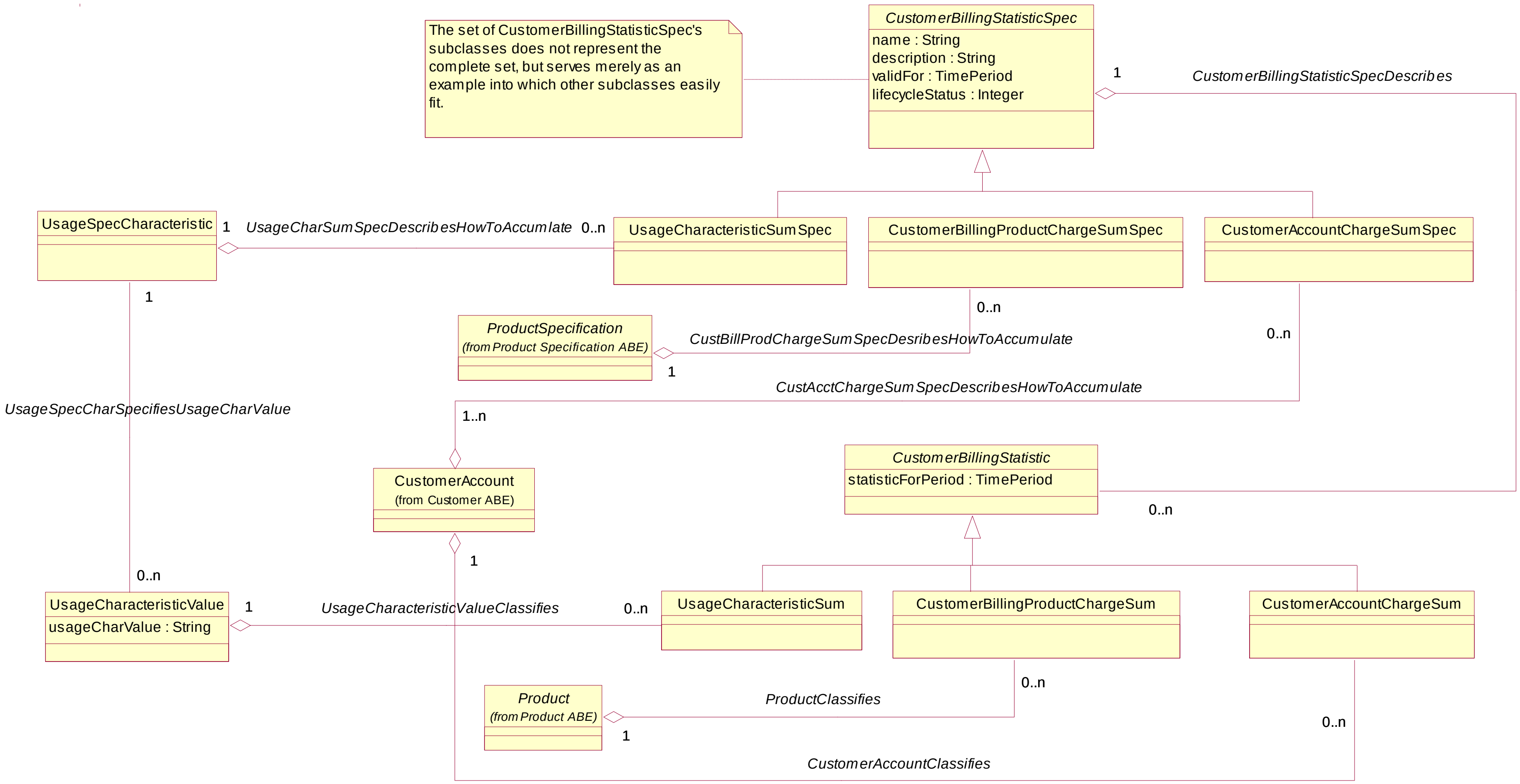

Figure C.15 – Customer Billing Statistic, Part 1

The CustomerBillingStatisticSpec is used to describe the statistic that needs to be collected. The UsageCharacteristicSumSpec describes the collection of usage characteristic statistic (for example, the total sum of the duration of voice calls in a month). Policy rules can be used to define, which statistic data are of the billing process interest, when they should be created and expired, how they should be calculated, and what actions should be triggered when statistic data meet certain conditions.

CustomerBillingStatistic entity instances are typically created and updated during rating and billing process. Since a quantity represented by CustomerBillingStatistic changes over time corresponding Balance entities are used to track its history as shown in the figure below.

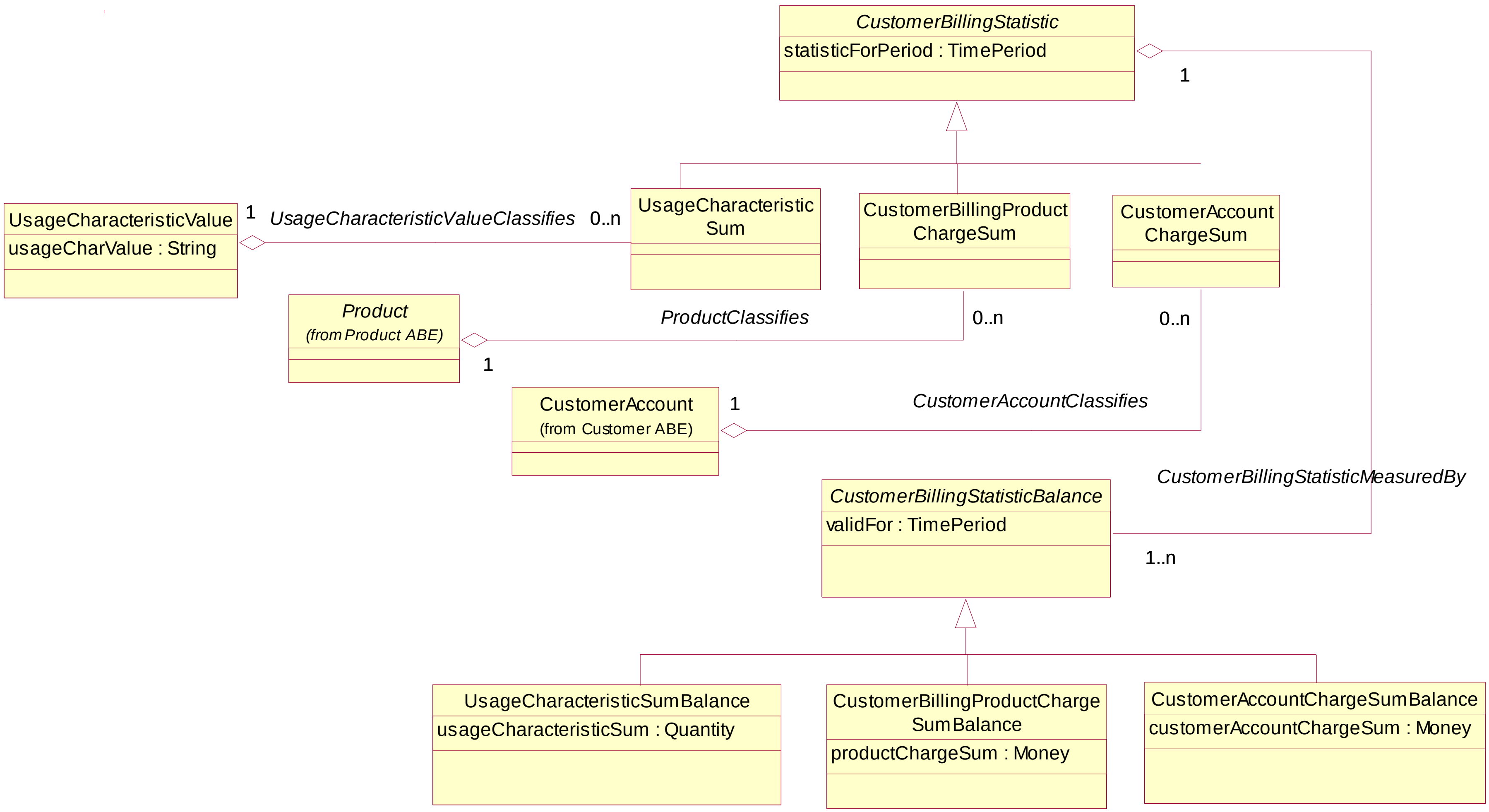

Figure C.16 – Customer Billing Statistic, Part 2

For example, for a certain product a billing system may track total number of kilobytes transferred over GPRS connection per quarter. The total sum is recorded by a billing system and stored in an instance of UsageCharacteristicSum entity. When a certain volume is exceeded a DiscountProductPriceAlteration instance may be generated entitling customer to a certain discount.

1.6.5. Customer Payment

This section is intended to define the scope of payment from the following perspectives. For more information about Payment refers to Party Payment in the Engaged Party Guide book.

- Why (the reason for payment)

  From the view of customer, a payment is usually applied to a bill or an invoice, maybe through a customer account, and applied to product and product usage in both prepaid and postpaid scenarios. The payment may represent a prepayment. The payment amount may be a credit or debit amount. The payment is eventually applied to a bill or FinanicalCharge/AppliedCustomerBillingRate.

- How (the methods and plan for payment)

  Payment is implemented through different means, such as cash, check, bank card, third party, account balance and loyalty burn, and so forth. A payment is applied to several bills or FinancialCharge/AppliedCustomerBillingRate according to the priority defined by payment plan.

- When

  In case of auto pay or receipt of customer payment

- Who

  Payee and Payer, the third party as an agency

- What (customer payment scope)

  Involve all the above info, which is described in Figure C.17 – Customer Payment Scope below.

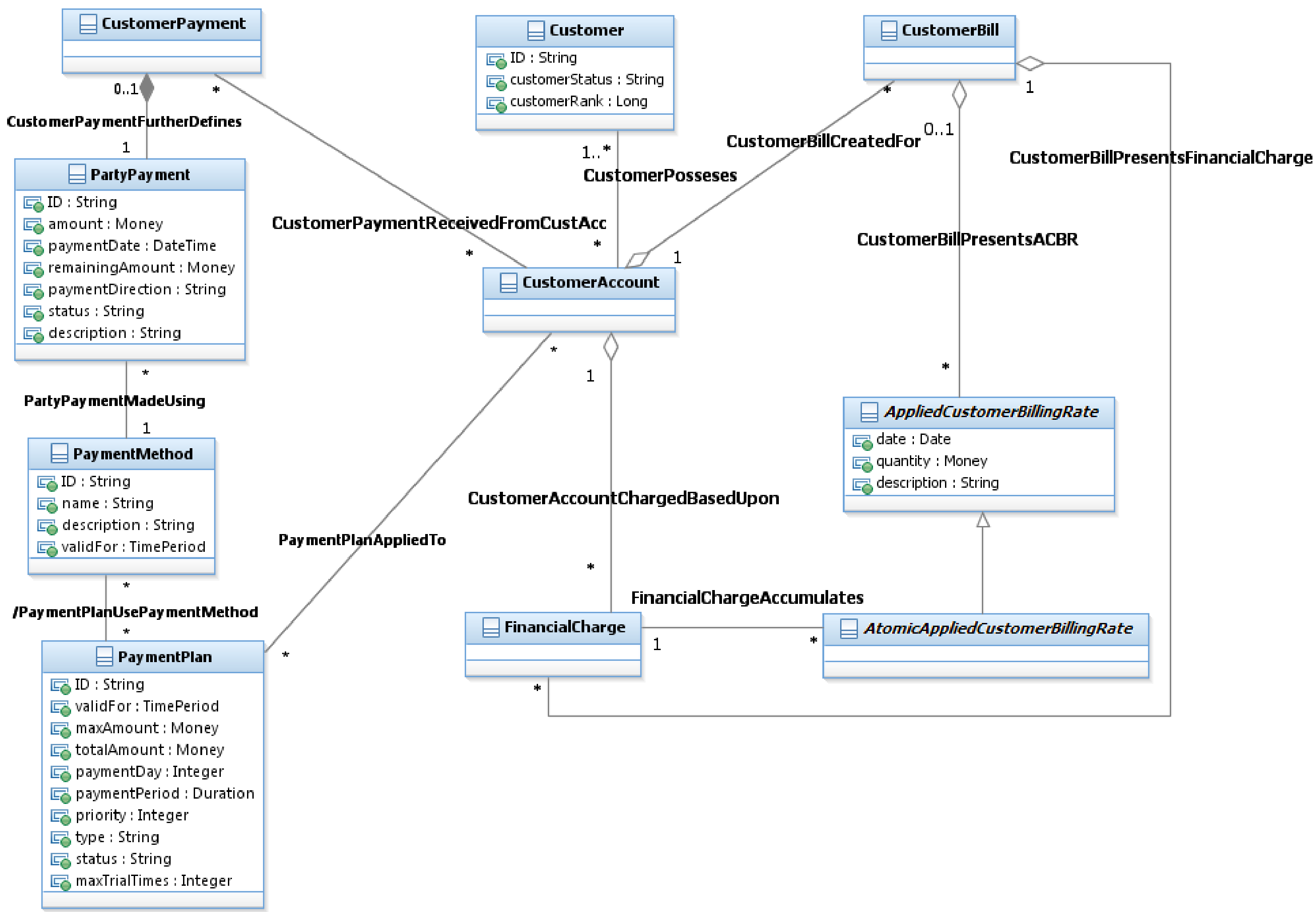

Figure C.17 – Customer Payment Scope

The payment model presented here is intended to support a converged payment model for both prepaid and postpaid conditions. Other requirements include:

- Handling prepaid customers in cases where there is no bill. So Payment could be related to FinanicalCharge/AppliedCustomerBillingRate and not to a CustomerBill.
- Handling a payment that is applied to multiple bills
- Handling underpayments and overpayments.

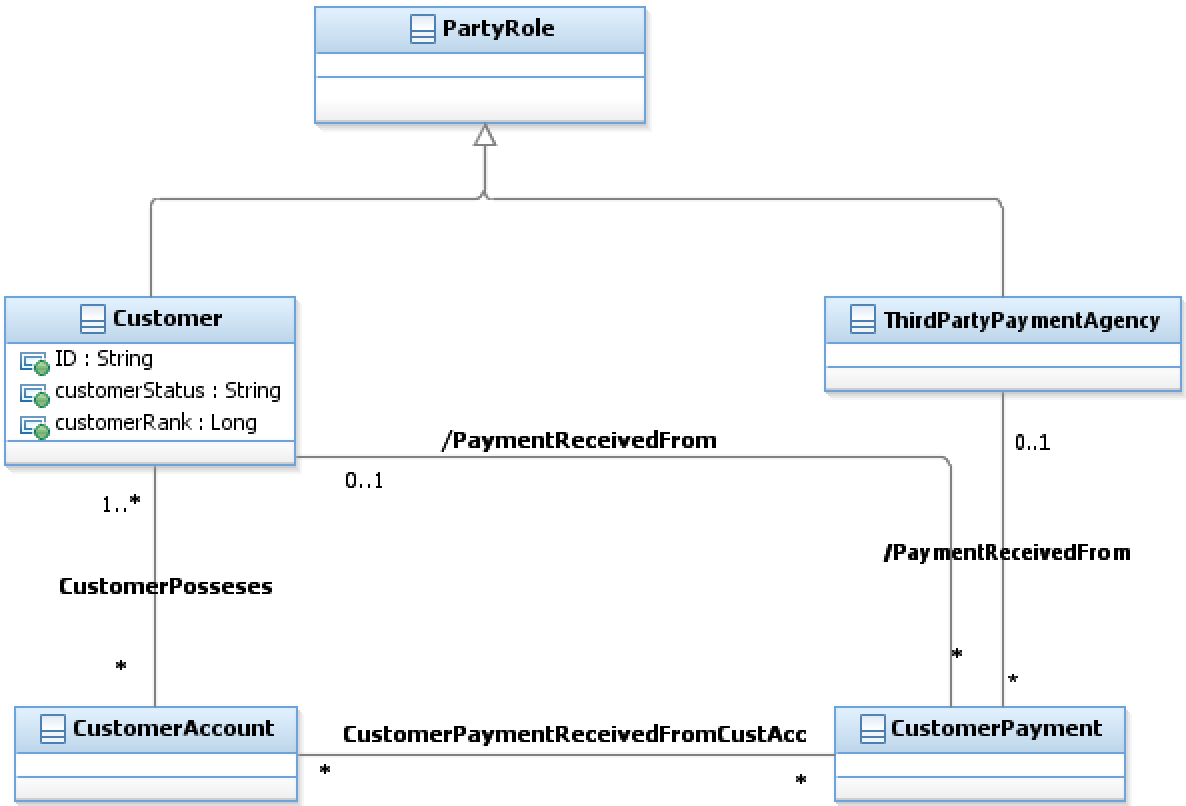

Figure C.18 – Payment's Associations with Roles

Customer and ThirdPartyAgency play a role as payee. Payment is received for one or more CustomerAccounts.

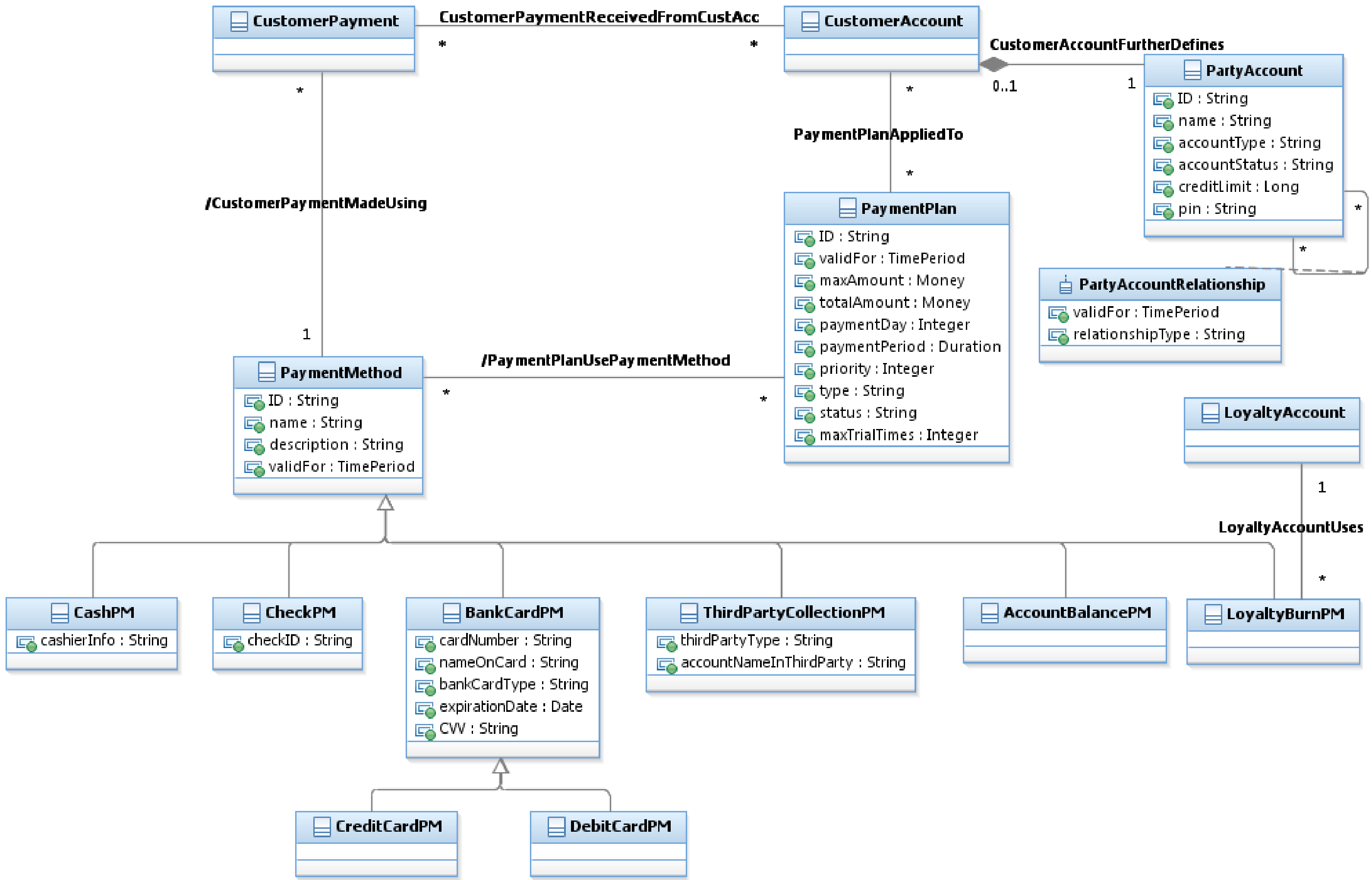

Figure C.19 – Payment Method and Plan

The means of payment is classified by PaymentMethod. For example, payment by cash, check, bank card (credit card or debit card), third party (like bank transfer, payment tool), account balance and loyalty burn. PaymentMethod has a generalization of the subclasses shown in Figure C.19 – Payment Method and Plan, which could be used by PaymentPlan. In case of auto payment, PaymentPlan will use PaymentMethod and the information from CustomerAccount to obtain the money and then apply them based on prearrangement, such as priority of bills based on payment due date.

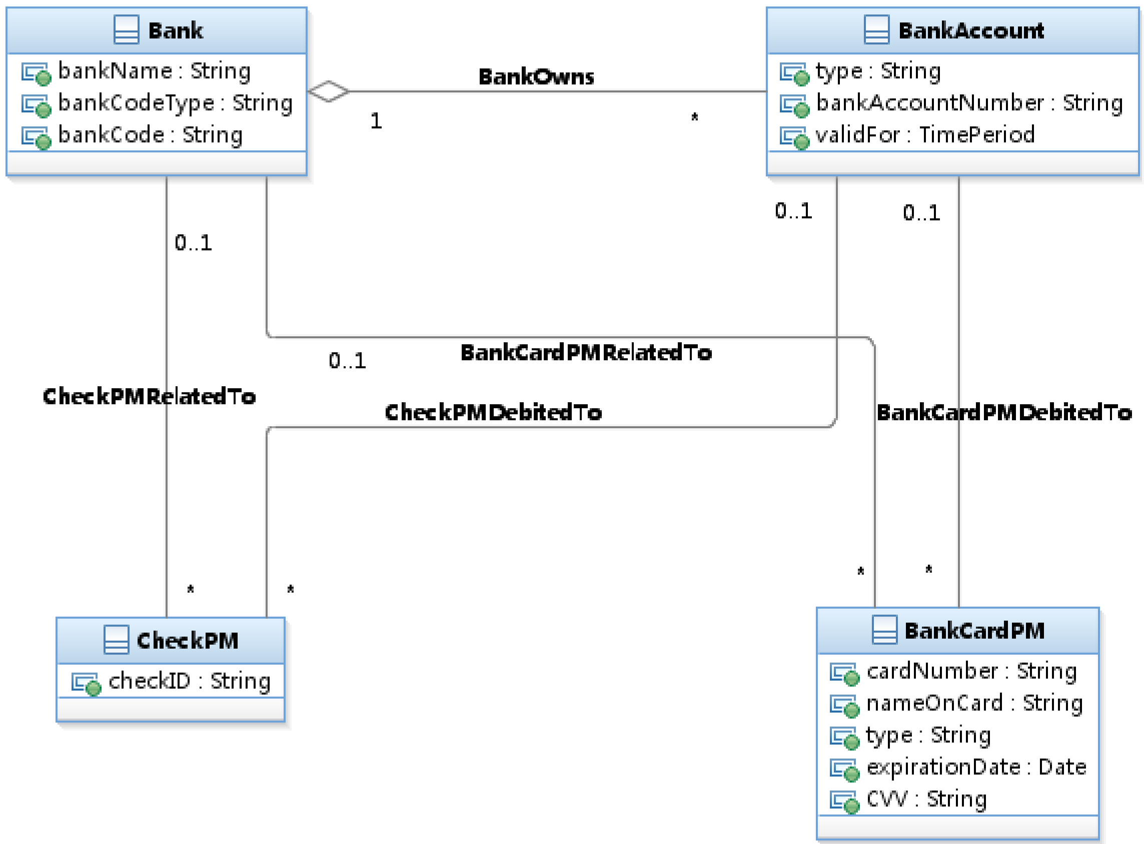

Figure C.20 – Payment Method and Bank

The relationship between PaymentMethod and Bank related entities is shown in Figure C.20 – Payment Method and Bank. Here the related subclasses of PaymentMethod are CheckPM and BankCardPM which has two subclasses of CreditCardPM and DebitCardPM.

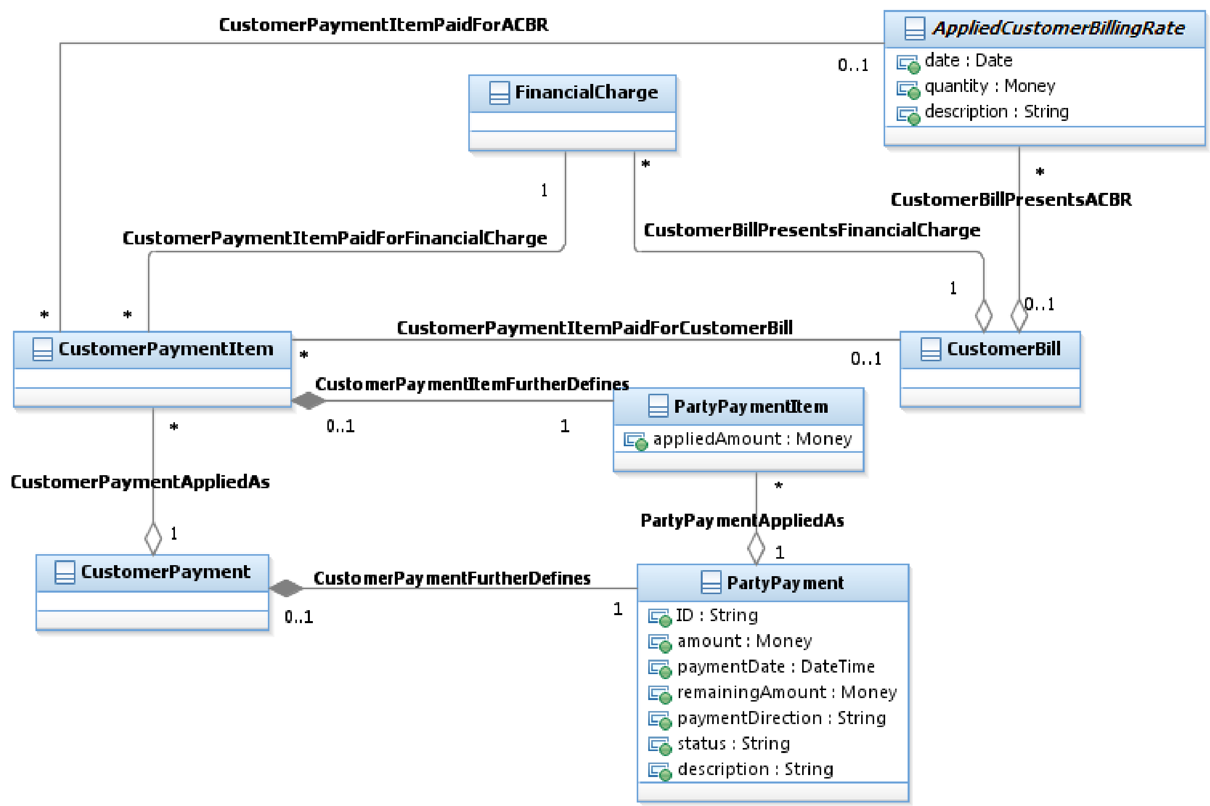

Figure C.21 – Applying Payment

As shown in Figure C.21 – Applying Payment, Payments can be applied to the CustomerBill and FinancialCharge/AppliedCustomerBillingRate.

1.6.6. Dunning

The Dunning ABE is part of the Customer Bill Collection ABE.

This section is intended to define the scope of information used by the dunning process.

More precisely, this section describes information produced and consumed by the level 3 business processes from eTOM

- 1.1.1.11.3 Manage Customer Debt Collection
- and 1.1.1.1.14 Support Bill Payments & Receivable Management

The following Figure C.22 – Dunning Scenario and Case presents the dunning scenario and the corresponding cases.

Each provider/operator has to define its dunning strategy. This strategy is described by DunningScenario. Each DunningScenario contains the dunning rules to apply to a case, the order in which applying them, the events that trigger dunning rules evaluation, the actions that must be done, and so forth.

The DunningCaseRules are specified by DunningRule that is a type of PolicyRule.

This allows specifying for each dunning rule:

- the events that trigger the evaluation of the rule (Ex: due debt not paid after 8 days
- conditions that have to be evaluated (Ex: no bill dispute in progress)
- and actions that have to be done (Ex: send a dunning letter).

A DunningScenario is assigned to a CustomerAccount, depending on the Holder quality (Ex: a more tolerant DunningScenario is assigned to VIPs).

If no specific DunningScenario is assigned to a CustomerAccount, a default scenario is applied.

As soon as an event triggering a DunningRule occurs on a CustomerAccount, a DunningCase is created according to the DunningScenario assigned to the CustomerAccount.

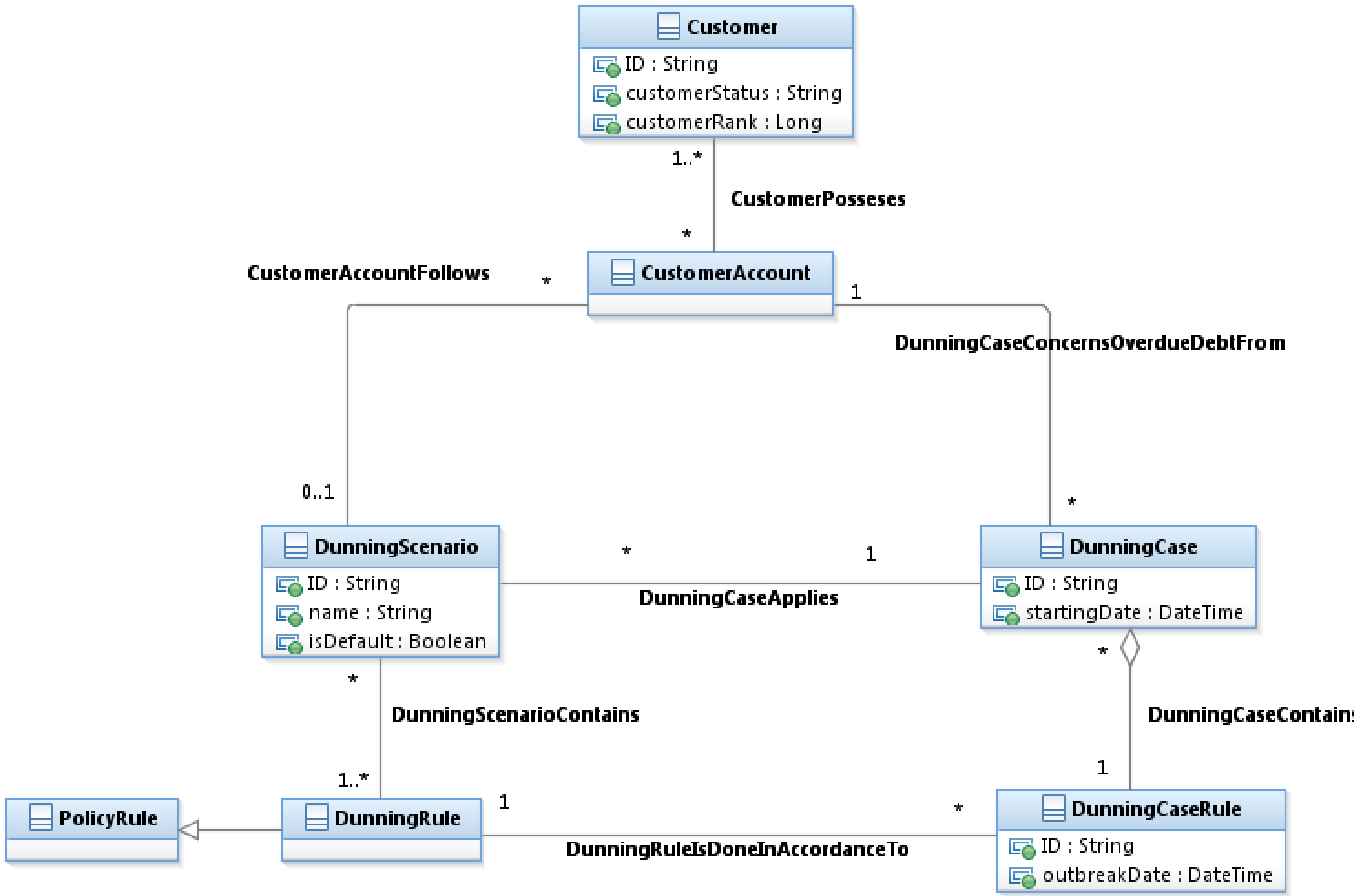

Figure C.22 – Dunning Scenario and Case

The following example illustrates the use of the dunning model and, more specifically, the inheritance from PolicyRule.

In this example, the provider/operator has defined a default Dunning scenario for Mass Market.

The Rules 1 and 1b use the same PolicyEvent and the same PolicyAction but their conditions are different. Only one of these two rules may be applied in a specific context.

In the Rules 2 and 2b

- the PolicyEvents and the PolicyAction are the same
- but the PolicyConditions are different: the amount of the due debt conditioning the action is only 20€ for a bad debtor and 100€ for a good payer.

The Rules 13 and 13b trigger two PolicyActions if the condition is true.

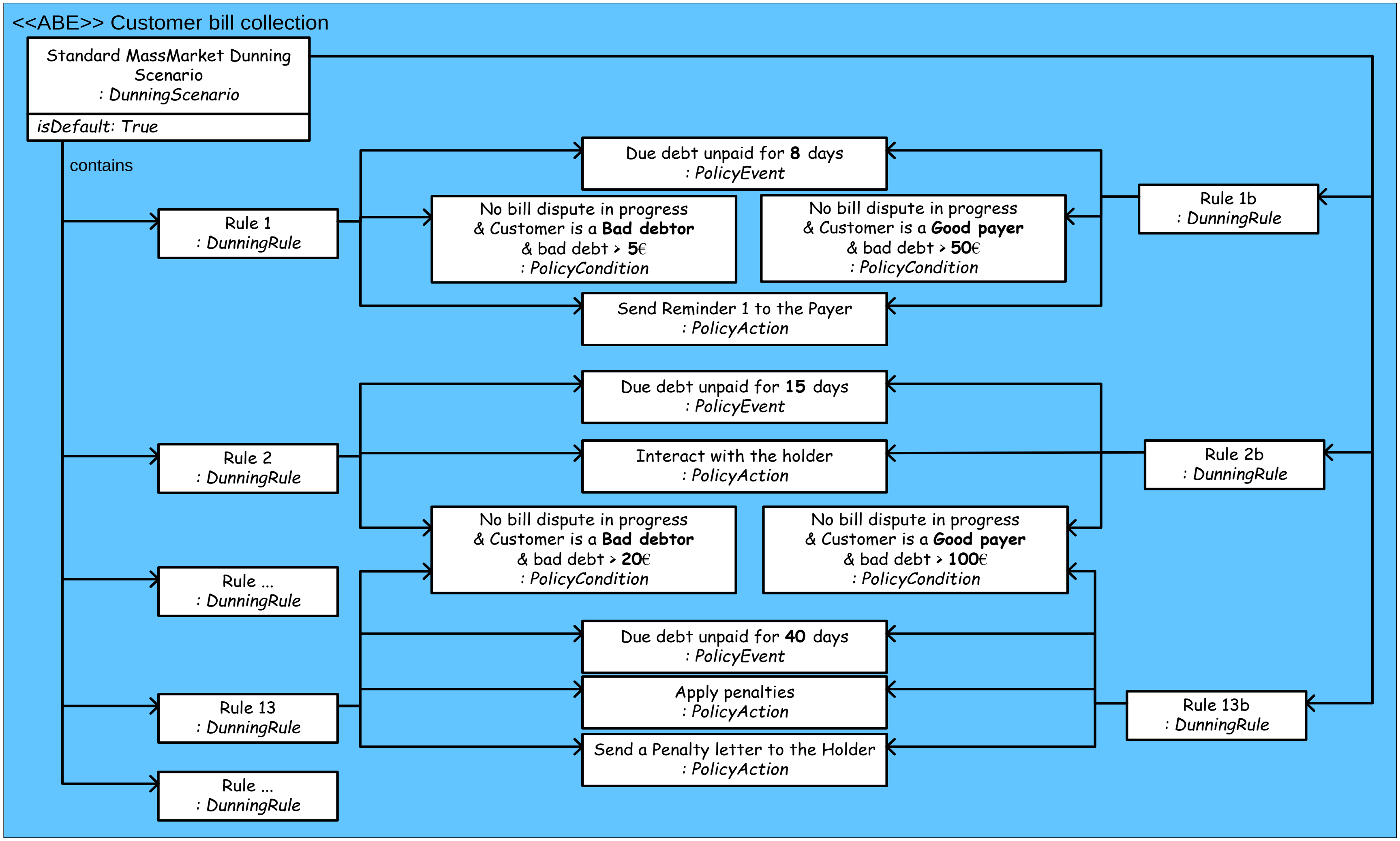

Figure CE.02 – Dunning Scenario Illustration

When a dunning process is triggered, it will first create a SoftDunning. At this stage the provider/operator starts to get in touch with the payer. And without result, we get in touch with the holder.

Finally, after restriction measures, the entire holder's contracts are removed and a closing invoice is produced.

Once the closing invoice is produced, a HardDunning starts.

According to the law, several interactions will be done with the holder and finally a DunningWriteOff is credited.

Each time DunningRule events occur, a DunningCaseRule is triggered to apply it.

If the condition specified by the DunningRule is true, then one or several actions are done.

The different types of dunning actions are the following:

- One type of dunning action may be to interact with the customer to ask for a CustomerPayment or negotiate a PaymentPlan;
- Another type of dunning action may be to apply penalty for overdue debt (AppliedCustomerPenaltyCharge that is a type of Charge);
- Another type of dunning action may be to create a CustomerOrder that will apply a limitation Product on the customer's existing Products (Ex: limit to national calls) and even a CustomerOrder that will remove all the Customer's Products;
- When all the Customer's Products are removed a closing CustomerBill is produced, including the rented material goods price;
- The last action done by the Dunning process is to produce a DunningWriteOff.

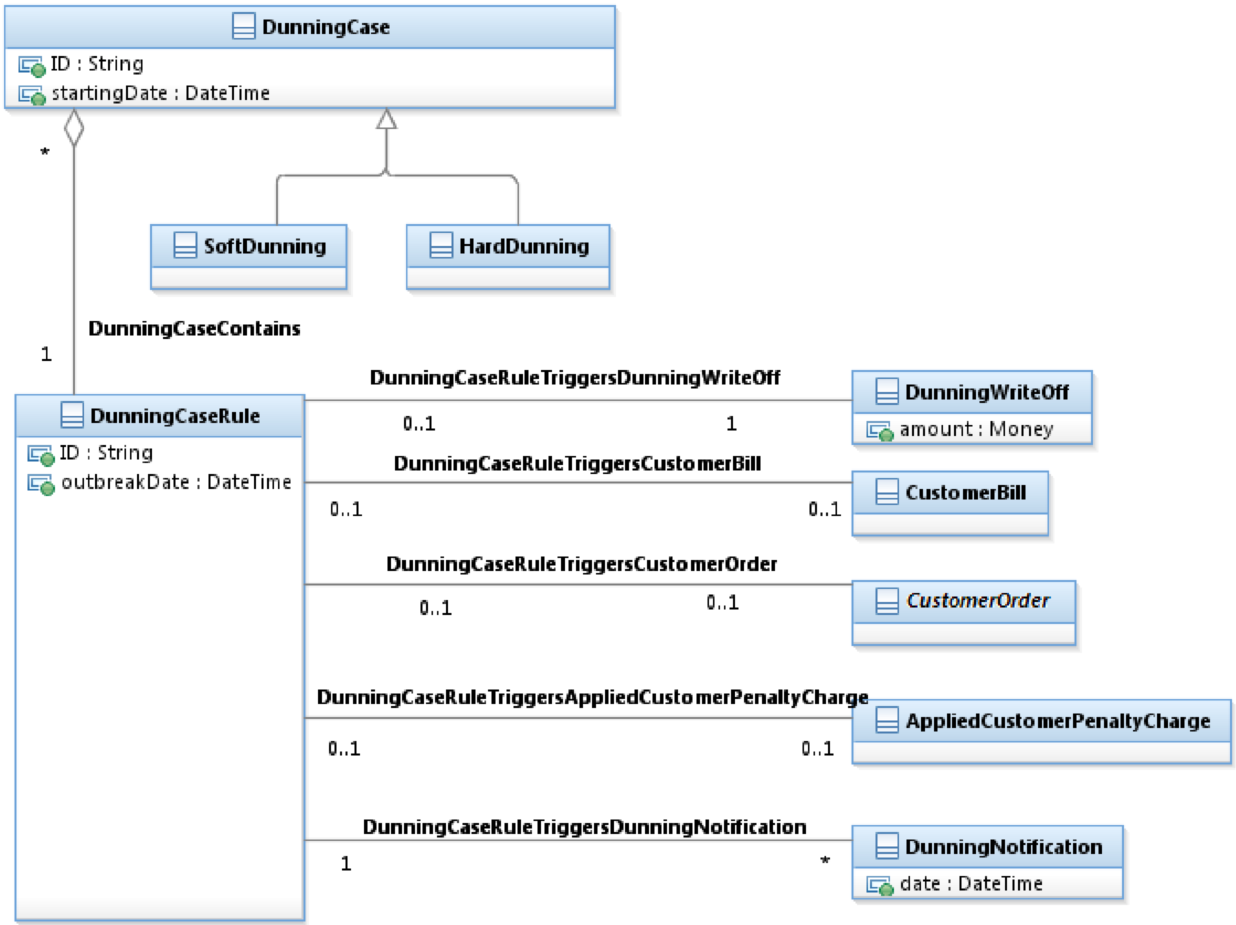

Figure C.23 – Dunning Result

The following example illustrates the application of a dunning scenario for a specific customer.

In this example, John Smith has a contract with the provider/operator. Therefore, he is liable for eventual debt that could be incurred on his CustomerAccount.
Mary Smith plays the role of Payer and pays the debt of John Smith' CustomerAccount.

As no specific dunning scenario has been assigned to John Smith' CustomerAccount, it follows the Standard MassMarket Dunning Scenario.
As Mary Smith didn't pay the due debt of the CustomerBill for 8 days, a DunningCase is created starting by the first step.
In accordance with the DunningRule, a reminder letter is sent to Mary Smith (the Payer).
As no payment has been received from Mary Smith, the second step of the Dunning Scenario is triggered. The CSR (Customer Service Representative) calls Mary Smith.
And hopefully, the interaction with Mary Smith led to a Payment covering the complete debt.

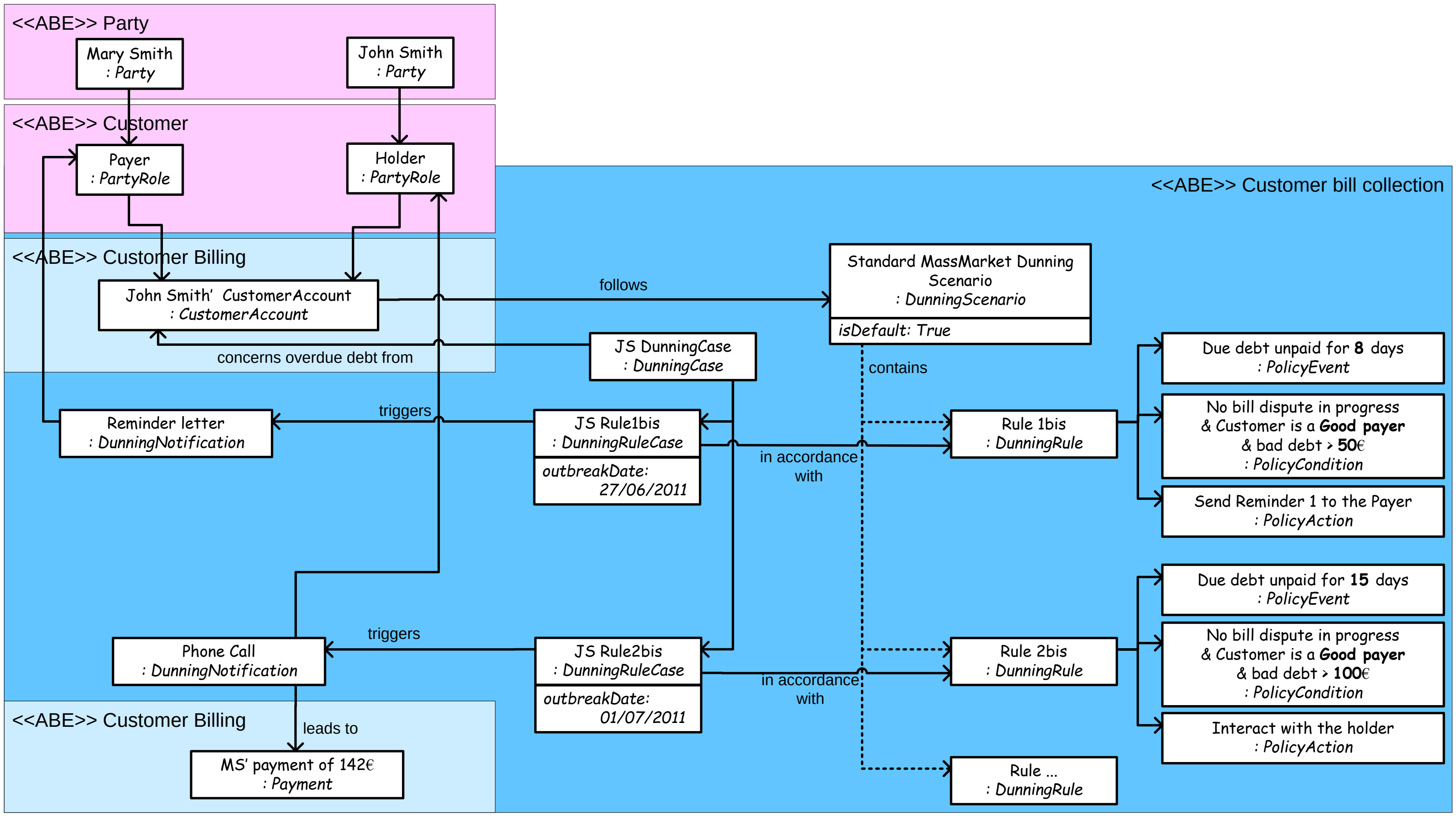

Figure CE.03 – Dunning Results Illustration

1.6.7. Customer Account Balance

Balance (or balances) represents an amount, usually with some monetary value, usually an aggregation, that indicates an aspect of liability (often financial) between the service provider and the account holder (usually a customer). This amount can be monetary amount, or non-monetary (such as free minutes, or amount of loyalty points), can represent credit that the CSP holds for the customer (for example pre-paid balance) or debt that the customer owes the CSP (as is the normal case for post-paid A/R balance). Balance can be limited by regulation (for example, in most countries pre-paid balance cannot be negative by regulation) or by business rules (credit limit on post-paid balance based on customer credit profile) and so on.

Balance, by nature, is a temporal field and in many cases it is important to capture not only the value of the balance at a specific point in time but also the change of the value with time. In many cases systems will need to be able to answer a query such as "what was the post-paid customer account balance for account XXX on date YYY".

Customer account balance can be used as the source of payment and/or as the target of payment. In some cases payments are designated to a specific balance (for example in case of top-up of pre-paid balance) and in some cases the designation is automatic (for example when the customer pays a post-paid bill). All these cases are captured by one concept (customer account balance) with some differentiation.

From the above it is clear that each CustomerAccount may contain more than one balance and therefore, according to SID guidelines CustomerAccountBalance is modeled as an entity with 1-* association to CustomerAccount.

Below is some explanation how monetary balances are associated with payments. While monetary balances are very important, we should remember that the generic case includes non-monetary balances as well which are more loosely related to payments.

Acting as customer account balance management, Customer Account Balance ABE is related tightly to customer payment and other ABEs especially in charging domain. It is intended to support a converged model applicable for both prepaid and postpaid cases.

The prepaid customer needs CustomerAccountBalance in recharging for storage. The payment model also need to handle prepaid customers in cases where there is no bill and payment goes directly into CustomerAccountBalance. Then CustomerAccountBalance amount could be applied to AppliedCustomerBillingRate which might be related to certain product usage event. From above, the prepaid customer needs CustomerAccountBalance both in recharging and payment.

Actually in case of overpayment or refund for prize, CustomerAccountBalance is the temporary method for storage to postpaid customer. From the view of postpaid customer, a payment is typically applied to a bill or an invoice, and indirectly applied to product and product usage. CustomerAccountBalance could also be involved in payment as a payment method/source to a bill or an invoice. So the postpaid customer also needs CustomerAccountBalance both in refunding and payment.

As shown in the Figure C.19, CustomerAccountBalance (working as AccountBalancePM) could be taken as a payment method/source for certain payment amount.

Customer can have multiple product charges, and CustomerAccountBalance can be shared between them, and used to pay product charges for other customers.

A payment may be applied to several bills or AppliedCustomerBillingRates according to the priority defined by payment plan. Before the bill or AppliedCustomerBillingRate is calculated, we need to associate product charges to certain CustomerAccount(s), by which product usage are paid because bill or AppliedCustomerBillingRate is in aggregation relationship with certain CustomerAccount(s). One CustomerAccount may contain many CustomerAccountBalances which could be associated to and pay for different product charges. Besides designating product charges to certain CustomerAccount(s), we need also designate product charges to certain CustomerAccountBalance(s) directly or indirectly. These designation relationships help to aggregate AtomicAppliedCustomerBillingRate(s) for product usage as certain financial charge and this financial charge is further associated with certain CustomerAccountBalance(s).

Broad scope for CustomerAccountBalance is shown as Figure C.24 below.

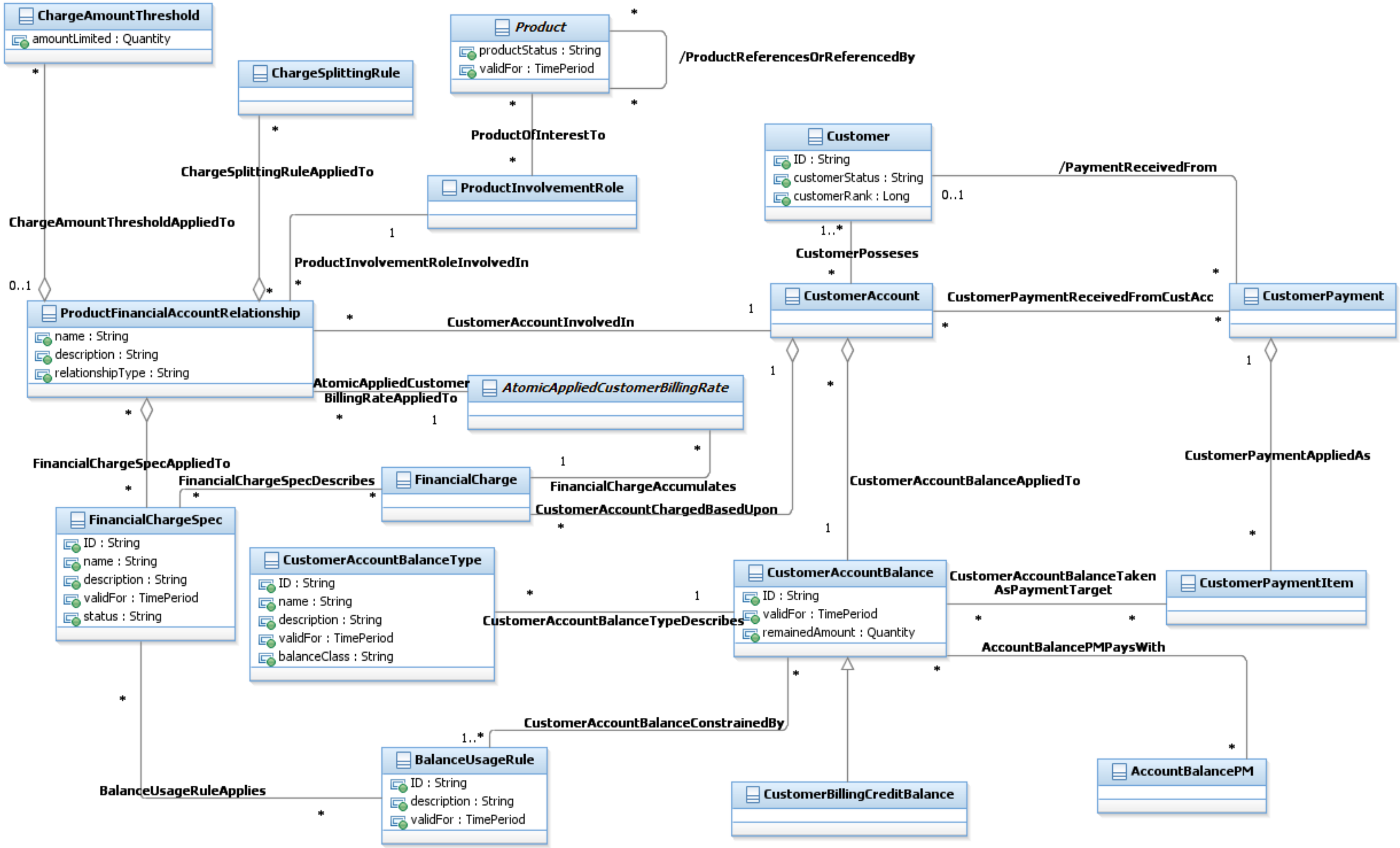

Figure C.24 – Customer Account Balance overview

Three key associated areas are product charge relationship area, core balance area and payment activity area.

Product charge relationship area contains relationships between product charge and CustomerAccount and relationships between product charge and financial charge specification. Product charge relationship helps to relate product charge to certain CustomerAccount(s) and carries out charging activities which result in possible financial charge.

Core balance area contains CustomerAccountBalance management entity and relates certain CustomerAccountBalance(s) with calculated amount waiting to be paid after charging process.
Payment activity area contains payment plan and carries out payment activities. Payment plan is associated with possible payment methods/sources and all kinds of payment target with different priority.
Product charge relationship area is as shown in the Figure C.25 below.

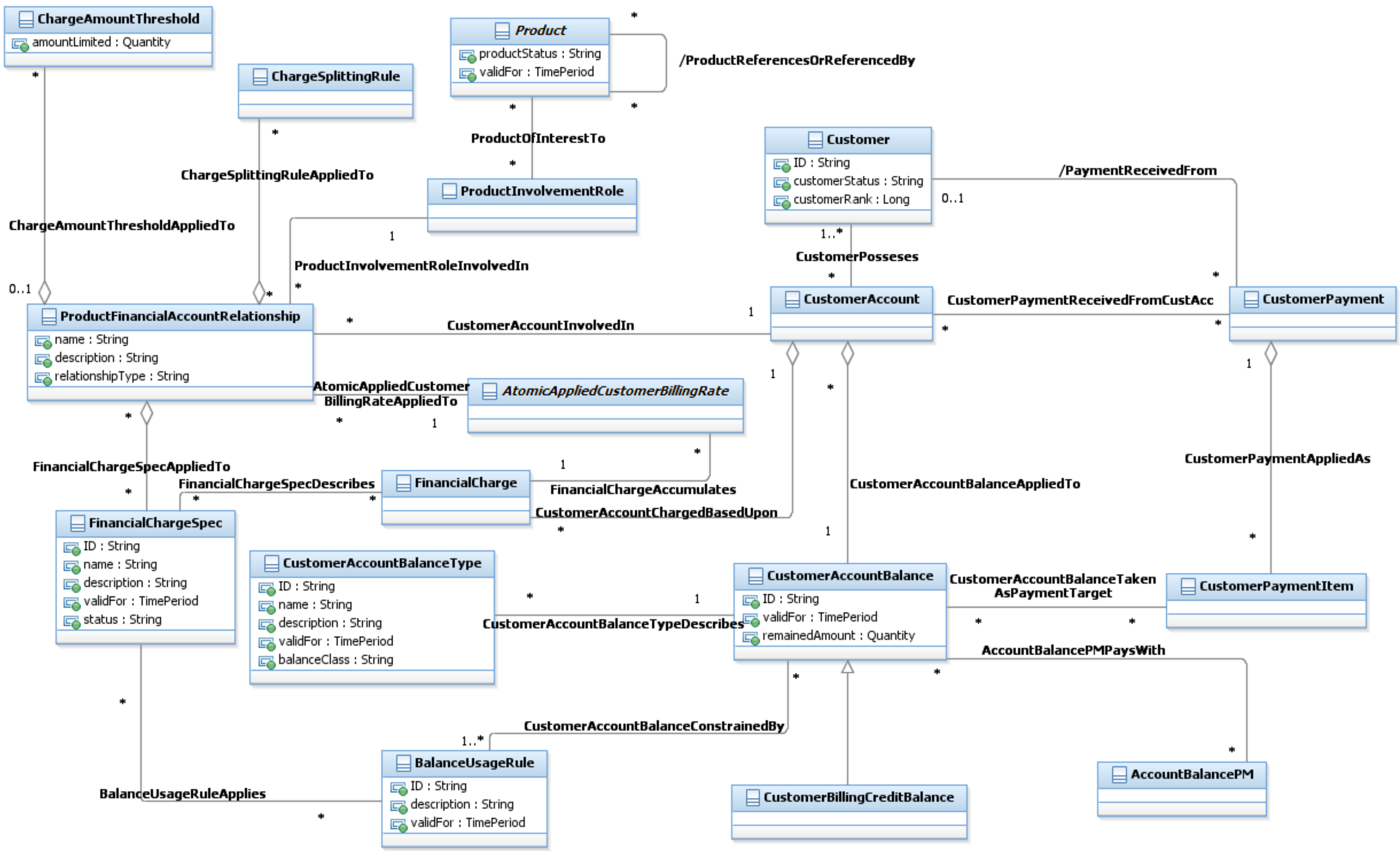

Figure C.25 – Customer Account Balance, Part 1

ProductChargeAccountRelationship entity relates product charge to one or more CustomerAccount(s) via which associated charging information is maintained. ProductChargeAccountRelationship uses rule information(from ChargeSplittingRule and ChargeAmountThreshold) and charge information held by FinancialChargeSpec and AtomicAppliedCustomerBillingRate to decide the charge relationship, such as which charge from one product usage should be directed to which CustomerAccount(s) and the charge amount. It is different from old FCAccumulatesAACBR where the CustomerAccount which pays for the charge was already decided.

ChargeAmountThreshold constrains relationship between product charges and CustomerAccount(s) by specifying a limited amount beyond which product charges will not be affordable by associated CustomerAccount.

ChargeSplittingRule is charge-level/account-level rule information describing which product charge should be directed to which CustomerAccounts. It can base on time and/or percentage. For example, if A gets a phone from the company and A pays 10% of the charges while the company pays 90% of the charges. For another example, the company pays all charges during the working time and the employee pays all charges out of the working time.

FinancialChargeSpec defines one specification for certain kind(s) of FinancialCharge. Instance of FinancialChargeSpec could be like long-distance call FinancialChargeSpec in contrast with local call FinancialChargeSpec. Based on FinancialChargeSpec, product charges could be directed to different CustomerAccounts. For example, one enterprise customer only pays for the local call fee of its employees which means its employees need to pay long-distance call fee via their own CustomerAccounts. Moreover, recorded in ProductChargeAccountRelationship, Product or Product component is in align with certain FinancialChargeSpec which means charging fee for this Product or Product component will be aggregated as certain associated FinancialCharge under certain CustomerAccount(s). One FinancialChargeSpec can be used in many different ProductChargeAccountRelationships instances. Reversely one ProductChargeAccountRelationship can use several different FinancialChargeSpecs. For example, ProductInvolvementRole is a long-distance call user. Here FCS might be one long-distance call FinancialChargeSpec. Moreover FinancialChargeSpec might also be several FinancialChargeSpecs(one might be local call FinancialChargeSpec, the other might be long-distance call FinancialChargeSpec). Here one or several FinancialChargeSpecs might be determined by how to design FCS and the government's regulation on how to charge on one product usage. One more example is that the charging policy is changed based on time (such as peak time charging policy and off-peak time charging policy) while the calling is going through the border between peak time and non-peak time. So the calling is associated with two FinancialChargeSpecs at last.

Core balance area is as shown in the Figure C.26 below.

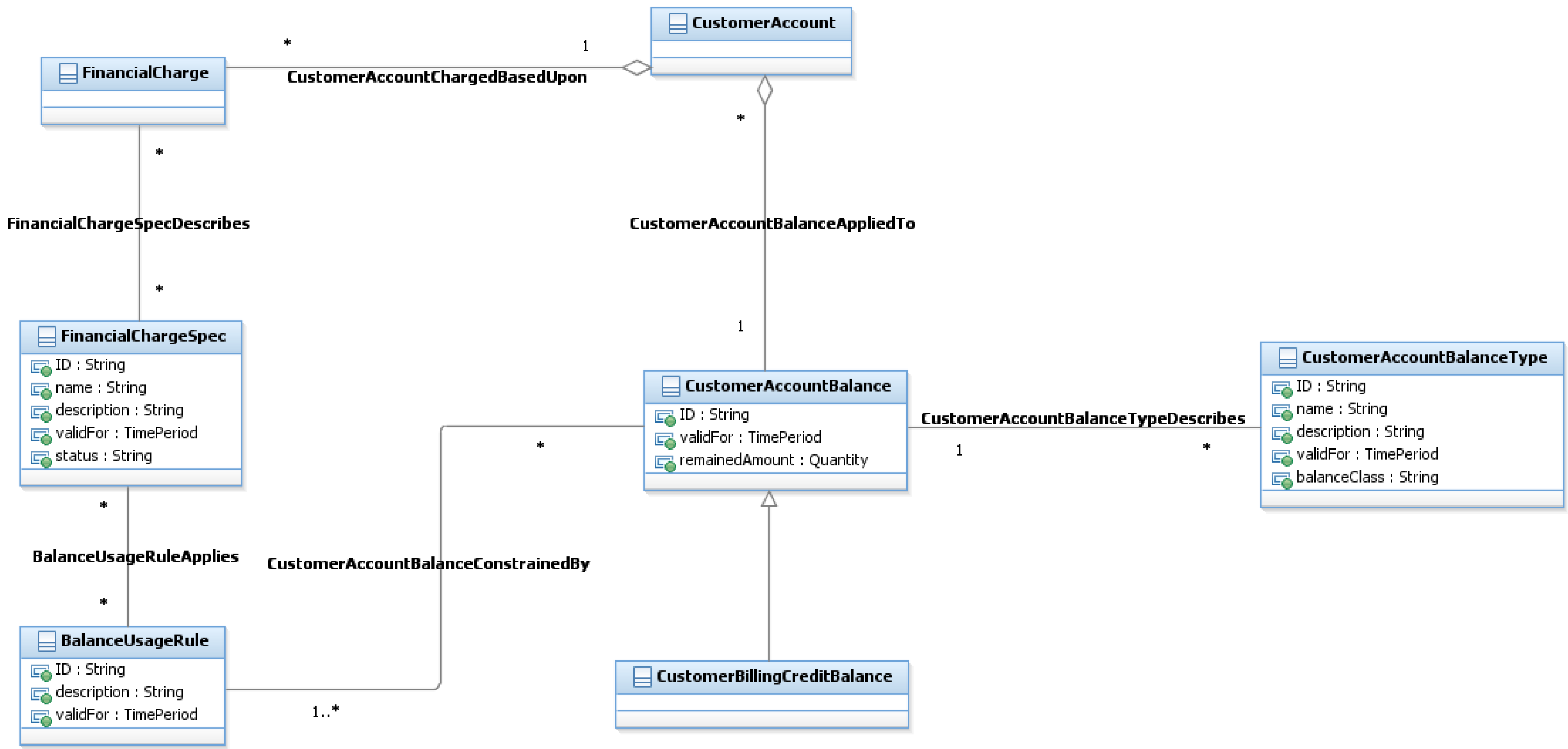

Figure C.26 – Customer Account Balance, Part 2

CustomerAccountBalance represents and tracks the amount remained or owed in certain CustomerAccount. The attribute remainedAmount is Quantity type. Here Quantity type represents both non-monetary balance and monetary balance. Non-monetary balance is applied to the service usage volume and monetary balance is applied to the monetary amount. CustomerBillingCreditBalance at the following section is a subclass of CustomerAccountBalance.

CustomerAccountBalanceTypeSpec describes and defines certain type of CustomerAccountBalance. Predefined CustomerAccountBalanceSpec will help to define different balances under CustomerAccount. The attribute accountBalanceType of CustomerAccountBalance and the attribute balanceType of CustomerAccountBalanceTypeSpec help to associate one or more CustomerAccountBalances with one CustomerAccountBalanceTypeSpec and the related CustomerAccountBalanceTypeSpec describes the type of CustomerAccountBalances.

BalanceUsageRule is balance-level rule information constraining the application of CustomerAccountBalance. It could describe the sharing rule, by which CustomerAccountBalance could be shared between different kinds of FinancialCharge which is related to different customer's product charge. It could emphasize associated CustomerAccountBalance is only applicable for certain usage, for example only applicable for local phone call or domestic flow or intra-PLMN SMS. FinancialChargeSpec could be applied to BalanceUsageRule to help to describe specification of FinancialCharge and relate them to the sharing information and the private usage information. BalanceUsageRule also includes priority and other kinds of rules limits the application of CustomerAccountBalance. For example, the refund SMS account balance should be used before monetary account balance when SMS is used.

Payment activity area is as shown in the Figure C.27 below.

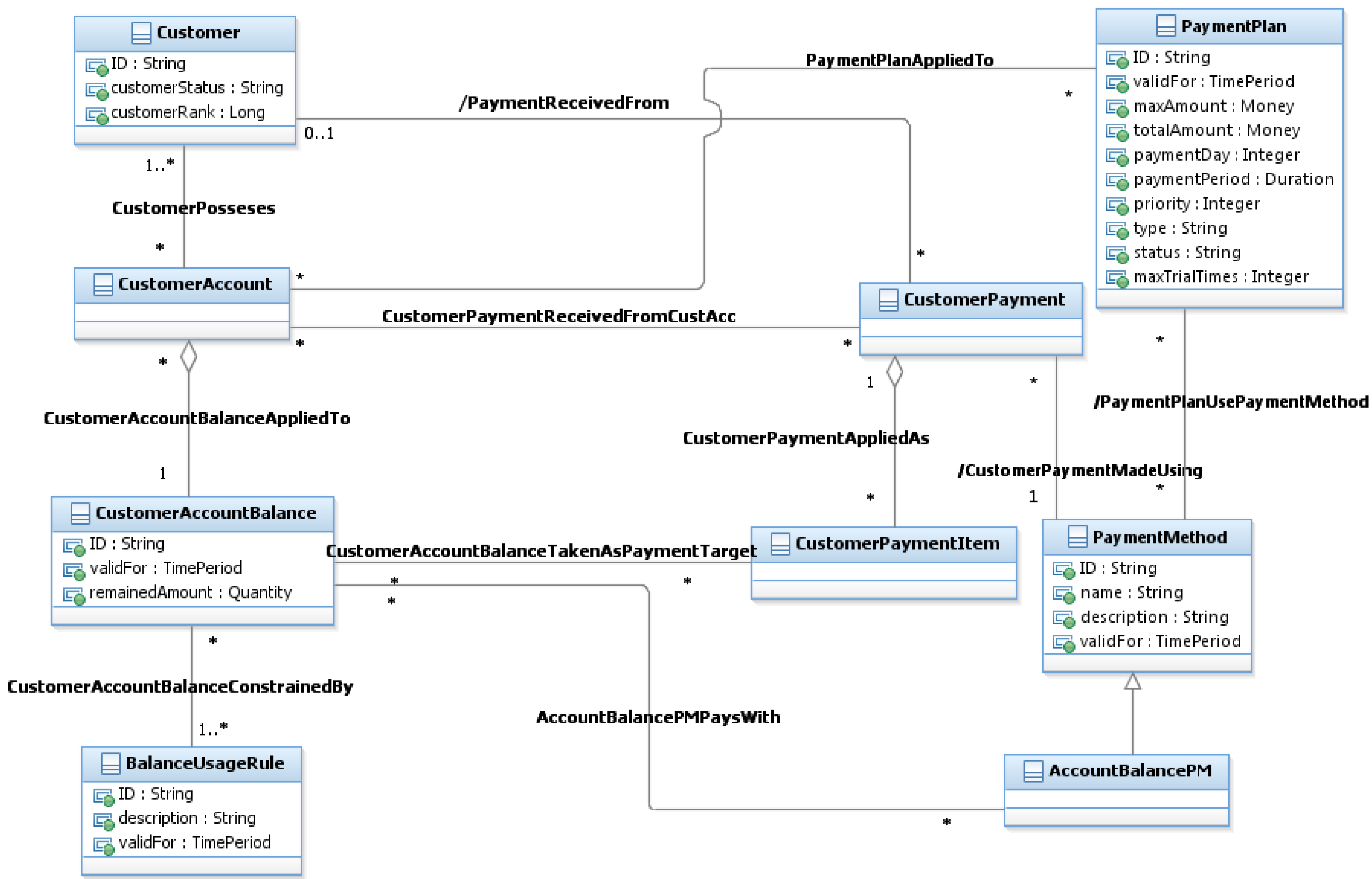

Figure C.27 – Customer Account Balance, Part 3

The payment model presented here is intended to support a converged payment model for both prepaid and postpaid condition. PaymentPlan could trigger CustomerPayment activities after Bill/AppliedCustomerBillingRate is calculated out. AccountBalancePM defines possible payment method using CustomerAccountBalance as payment source. When CustomerAccountBalance is the payment source, BalanceUsageRule constraints carrying out of payment activities.

Customer Billing Credit

The creation of CustomerBillingCredit instances is usually governed by ProductOfferingPriceRules. The ProductOfferingPriceRule or some other business logic can grant a certain credit to a customer account. When a part of the credit is applied during the billing process, the available credit is reduced. The CustomerBillingCredit and related business entities keep track of the credit. Because the balance of the CustomerBillingCredit changes over the time, the CustomerBillingCreditBalance entity represents the balance valid for a certain time period.

The CustomerDiscount business entity is a type of the CustomerBillingCredit. It keeps track of the available discount quantity for the associated DiscountProductPriceAlteration. For example, the DiscountProductPriceAlteration grants $100 discount on the Product charge. As the customer uses the service, the billing process is creating AppliedCustomerBillingCharge instances during each billing cycle, and the available discount amount is reduced. The billing process utilizes the CustomerDiscountBalance to keep track of the available discount (that has not been used yet).

The CustomerAllowanceBalance business entity similarly keeps track of the available allowance (that has not been used yet).

The figure below serves merely as an example of possible types of the CustomerBillingCredit that can be easily extended to keep track of other credits. Moreover CustomerBillingCreditBalance is a subclass of CustomerAccountBalance. CustomerBillingCreditBalance can be extended to keep track of other credit balance, not only non-monetary credit balance but also monetary credit balance.

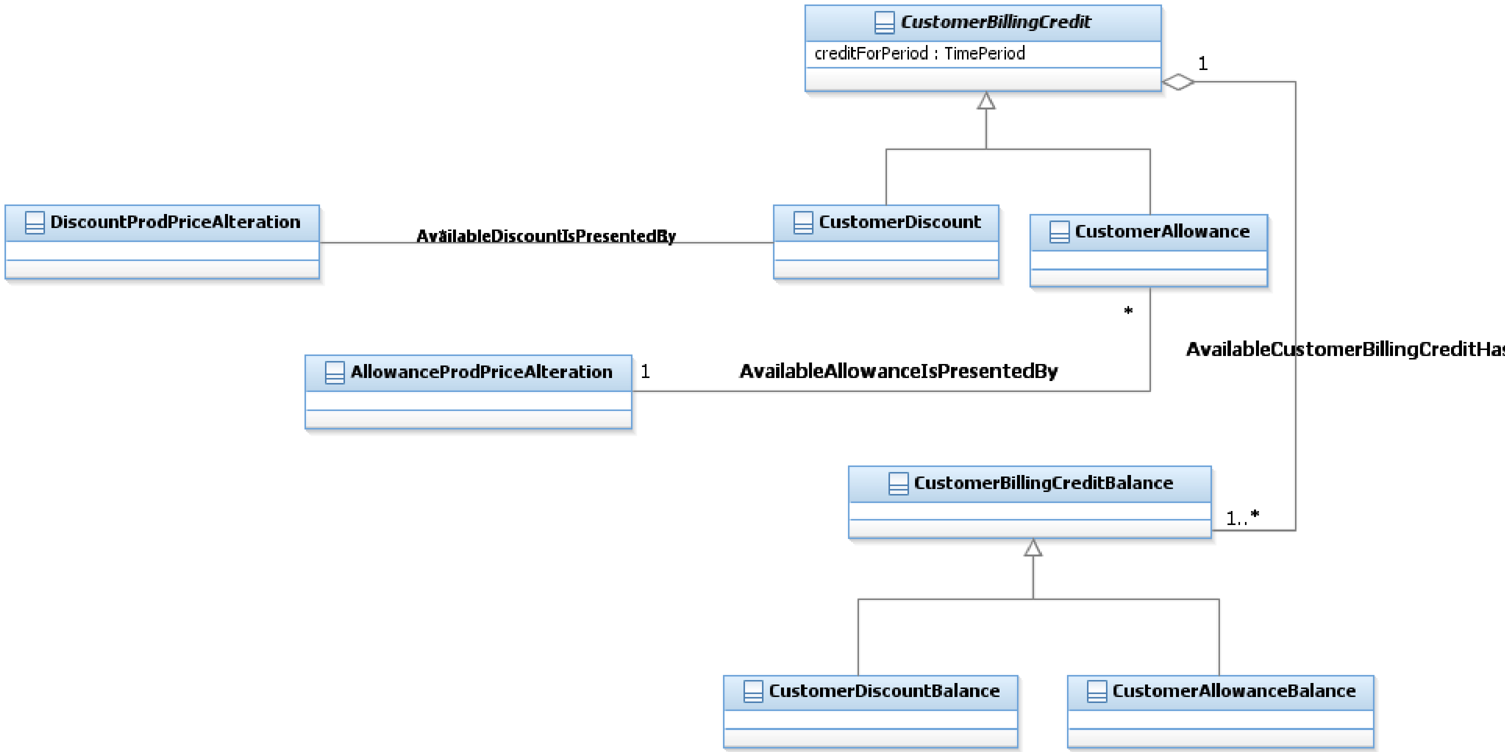

Figure C.28 – Customer Billing Credit

2. Administrative Appendix

This Appendix provides additional background material about the TM Forum and this document. In general, sections may be included or omitted as desired; however, a Document History must always be included.

2.1 About this document

This is a TM Forum Guidebook. The guidebook format is used when:

- The document lays out a 'core' part of TM Forum's approach to automating business processes. Such guidebooks would include the Telecom Operations Map and the Technology Integration Map, but not the detailed specifications that are developed in support of the approach.
- Information about TM Forum policy, or goals or programs is provided, such as the Strategic Plan or Operating Plan.
- Information about the marketplace is provided, as in the report on the size of the OSS market.

2.2 Document History

2.2.1 Version History

Version Number	Date	Modified by:	Purpose
0.1	Jan 2002		GB922 Addendum Template created from TMF407 v1.1 Guidebook template
0.5	May 2002		Initial member review version
1.0	June 2002		Formatted for Member Evaluation

Version Number	Date	Modified by:	Purpose
1.1	Oct 2002		Phase II updates
3.0	May 2003		Phase III updates, including updated association and attribute names.
5.0	June 2004		Updates based on member feedback.
5.1	August 2004		Synchronized with other Addenda
5.2	February 2005		Update based on member feedback
6.0a	April 2005		Update based on member feedback
6.0b	July 2005	John Reilly	Updated with Marand's Billing model
6.1	November 2005	Tina O'Sullivan	Converted to new template and corrected various administrative items.
6.2	November 2005	Tina O'Sullivan	Figure label
6.3	November 2005	Tina O'Sullivan	realigned a number of Figures
6.4		Tina O'Sullivan	Updated notice statement & document status
6.5	May 2009	Alicja Kawecki	Minor updates to reflect TM Forum Approved status
9.0	Jan 2010	Josh Salomon	Applied CR artf1869 - needed to change diagram C.12 to reflect entity name change
9.1	Apr 2010	Alicja Kawecki	Minor corrections for web posting and ME
9.2	Jun 2010	Alicja Kawecki	Updated Notice
9.3	Oct 2010	Alicja Kawecki	Updated to reflect TM Forum Approved status

Version Number	Date	Modified by:	Purpose
9.5	Nov 2010	Sihao Li	Updated with customer payment model proposed by Huawei
9.6	Mar 2011	Alicja Kawecki	Minor formatting corrections prior to web posting and ME
9.7	Sep 2011	Alicja Kawecki	Updated to reflect TM Forum Approved status
9.8	Jan 2012	Josh Salomon	Updated with FinancialCharge and DisputedAmount contributions (artf2441 and artf2647)
9.9	Mar 2012	Alicja Kawecki	Minor cosmetic corrections prior to web posting and ME
9.10	Oct 2012	Alicja Kawecki	Updated to reflect TM Forum Approved status of R12.0
10.0	July 2012	Cécile Ludwichowski	Updated with the Dunning ABE located in the Customer Bill Collection ABE
10.1	Oct 2012	Alicja Kawecki	Minor style/cosmetic corrections prior to web posting and Member Evaluation
10.2	April 2013	Josh Salomon	Implemented CR artf3324
10.3	May 2013	Alicja Kawecki	Rebranding, corrected copyright in footer
10.4	June 2013	John Reilly	Deleted duplicated ID attribute in DunningNotification. Removed duplicate name attribute from ThirdPartyPayeeAgency
10.5	Sep 2013	Josh Salomon	Added CustomerProblem ABE
10.5.1	Oct 2013	Alicja Kawecki	Updated cover, header and footer
10.5.2	Mar 2014	Alicja Kawecki	Updated cover, copyright year & Notice reflecting TM Forum Approved status

Version Number	Date	Modified by:	Purpose
10.5.3	Apr 2014	Yiling(Sammy) Liu	Added Customer Account Balance ABE
10.5.4	May 2014	Alicja Kawecki	Updated cover, header & Notice; minor formatting edits prior to posting
14.5.0	Oct 2014	Yiling(Sammy) Liu	Updated Customer Account Balance ABE and Customer Billing Credit ABE
15.0.0	Apr 2015	Yiling(Sammy) Liu	Implement CR artf2893 artf5365 artf5215 Implement CR artf5300 (Customer Payment Enhancement) Remove all Business Entity Definitions
15.0.1	May 2015	Alicja Kawecki	Updated cover, header; minor cosmetic corrections
15.5.0	Nov 2015	Cecile Ludwichowski	Updated with changes for EP Payment that impacts CustomerPayment Corrections for adding types on PaymentPlan's attribute
15.5.1	Nov 2015	Alicja Kawecki	Updated cover, minor cosmetic fixes prior to publishing
16.0.0	Apr 2016	Cécile Ludwichowski	Updated with changes for EP Order that impacts Customer Order.
16.0.1	25 May 2016	Alicja Kawecki	Minor cosmetic corrections prior to publication for Fx16
16.5.0	27-Oct-2016	Cécile Ludwichowski	Correct the Figure C.03 to remove a unidirectional relationship
16.5.1	18-Nov-2016	Alicja Kawecki	Minor cosmetic corrections prior to publication for Fx16.5
16.5.2	21-Feb-2017	Alicja Kawecki	Updated cover, header, footer and Notice to reflect TM Forum

Version Number	Date	Modified by:	Purpose
			Approved status; applied rebranding

2.2.2 Release History

Release Number	Date Modified	Modified by:	Description of changes
Release 6.0	31-Oct-2005	J. Reilly	
Release 9.0	3-Jan-2010	J. Salomon	Changed diagram C.12
Release 9.5	20-Nov-2010	Sihao Li	Updated with customer payment model proposed
Release 12.0	30-Jan-2012	Josh Salomon	Updated with FinancialCharge and DisputedAmount contributions
Release 12.5	5 Oct 2012	Cécile Ludwichowski	Updated with the Dunning ABE located in the Customer Bill Collection ABE
Release 13.0	24-Apr-2013	Josh Salomon John Reilly	Updated with onCycle and OffCycle Customer Bills. Deleted duplicated ID attribute in DunningNotification. Removed duplicate name attribute from ThirdPartyPayeeAgency.
Release 13.5	29-Sep-2013	Josh Salomon	Added CustomerProblem ABE
Release 14.0	31-Apr-2014	Yiling(Sammy) Liu Cécile Ludwichowski	Added Customer Account Balance ABE Added LoyatyBurnPM as a sub-class of

			PaymentMethod
Release 14.5	31-Oct-2014	Yiling(Sammy) Liu	Updated Customer Account Balance ABE and Customer Billing Credit ABE
Release 15.0.0	Apr 2015	Yiling(Sammy) Liu	Implement CR artf2893 artf5365 artf5215 Implement CR artf5300 (Customer Payment Enhancement) Remove all Business Entity Definitions
Release 15.5.0	9-Nov-2015	Cecile Ludwichowski	Updated with changes for EP Payment that impacts CustomerPayment Corrections for adding types on PaymentPlan's attribute
Release 16.0.0	27-Apr-16	Cécile Ludwichowski	Updated with changes for EP Order that impacts Customer Order.
Release 16.5.0	27-Oct-2016	Cécile Ludwichowski	Correct the Figure C.03 to remove a unidirectional relationship

2.3 Acknowledgments

This document was prepared by the members of the TM Forum Information Framework (SID) team.

The Shared Information/Data Model is a genuinely collaborative effort. The TM Forum would like to thank the following people for contributing their time and expertise to the production of this document. It is just not possible to recognize all the organizations and individuals that have contributed or influenced the introduction. We apologize to any person or organization we inadvertently missed in these acknowledgments.

Key individuals that reviewed, provided input, managed, and determined how to utilize inputs coming from all over the world, and really made this document happen were:

Name	**Affiliation**
Ian Best	TM Forum
Chris Hartley	Telstra
Helen Hepburn	BT
John Reilly	MetaSolv Software
Wayne Sigley	Telstra
John Strassner	Motorola
Dominik Roblek	Marand
Boris Cimperman	Marand
Tomaz Gornik	Marand
Bostjan Keber	Marand
Wayne Tackabury	Intelliden
Josh Salomon	Amdocs
Sihao Li	Huawei
Cécile Ludwichowski	Orange
Yiling (Sammy) Liu	Huawei
Jerry Zhu	Huawei